

B.Comp. Dissertation

**An End-to-end Framework for Knowledge Retrieval from Visual
Documents with Applications in Venture Capital**

By

Koh Quan Wei Ivan

Department of Information Systems and Analytics

School of Computing

National University of Singapore

2023/2024

B.Comp. Dissertation

**An End-to-end Framework for Knowledge Retrieval from Visual
Documents with Applications in Venture Capital**

By

Koh Quan Wei Ivan

Department of Information Systems and Analytics

School of Computing

National University of Singapore

2023/2024

Project No: H248030

Advisor: Assistant Professor Qiao Dandan

Deliverables:

Report: 1 Volume

Abstract

This dissertation presents an end-to-end framework for knowledge retrieval from visually rich documents, focusing on applications in the venture capital domain. The framework addresses the challenges faced by venture capital professionals in manually analysing pitch decks during the deal-sourcing process. It comprises three phases: Detection, Extraction, and Synthesis.

The Detection Phase introduces the IIIT-OSV-Charts dataset, a novel combination of the IIIT-AR-13K dataset and a proprietary dataset from Openspace Ventures. State-of-the-art YOLO object detection models are employed to accurately identify and localize chart instances within pitch decks. In the Extraction Phase, the Set-of-Marks prompting strategy is adapted for grounded zero-shot understanding of charts using large multimodal models. Relevant insights are extracted and stored in a vector database for efficient retrieval. The Synthesis Phase develops a Retrieval-Augmented Generation pipeline tailored to generate comprehensive responses to frequently asked questions crucial for decision-making. This pipeline is integrated with a user-friendly web application.

The framework is evaluated using a hybrid approach, demonstrating its effectiveness in extracting relevant information and generating accurate responses. Significant contributions include the creation of a novel dataset, the adaptation of advanced prompting strategies, and the development of a tailored pipeline. The framework has the potential to revolutionize the way venture capital professionals analyse and extract insights from visually rich documents, leading to more informed and efficient investment decision-making.

Subject Descriptors:

I.2.6 Learning

I.7.0 General Document and Text Processing

H.3.3 Information Search and Retrieval

Keywords:

Machine Learning, Image Analysis & Processing, Data Science and Business Analytics

Acknowledgement

I would like to express my deepest gratitude to my supervisor, Prof. Qiao Dandan, for her invaluable guidance, support, and encouragement throughout the entire process of writing this dissertation. Her expertise and insights have been instrumental in shaping the direction of this project and improving its overall quality. I am truly grateful for her dedication and the time she has invested in mentoring me.

I would also like to extend my sincere thanks to my co-supervisor, Prof. Zhao Yiliang, who is also the director of data science at Openspace Ventures. His support and connections within Openspace Ventures have been crucial in facilitating access to valuable resources and data that have greatly contributed to the success of this project.

I am immensely grateful to Openspace Ventures for their collaboration and the personnel who participated in my surveys. Their insights and feedback have been essential in refining my ideas and expanding my understanding of the venture capital domain.

I would like to express my sincere appreciation to the NUS AI Lab (NUSAIL) for providing me with a grant for this final-year dissertation. The funds have been instrumental in supporting my research and enabling me to contribute to the advancement of AI applications. I hope that I have utilised the resources effectively to make a meaningful impact in the field.

I would like to thank my friends and family for their unwavering support and encouragement throughout my academic journey.

Lastly, I would like to acknowledge the contributions of all the individuals who have directly or indirectly supported me in completing this dissertation. Your assistance and support have been greatly appreciated.

Table of Contents

ABSTRACT	III
ACKNOWLEDGEMENT	IV
TABLE OF CONTENTS	V
1. INTRODUCTION	1
1.1 PROBLEM AND MOTIVATION.....	1
1.1.1 <i>An Emphasis on Pitch Deck Documents</i>	2
1.1.2 <i>Importance of Charts in VC Deal Sourcing</i>	3
1.1.3 <i>Frequently Asked Questions (FAQs) in VC Deal Sourcing</i>	3
1.2. OBJECTIVES AND SOLUTIONS.....	5
2. LITERATURE REVIEW	7
2.1 DOCUMENT IMAGE TRANSFORMER (DiT) AND YOLO (YOU ONLY LOOK ONCE) IN DOCUMENT IMAGE ANALYSIS.....	7
2.2 CHARTQA AND EXTRACTING INFORMATION FROM CHARTS.....	9
2.2.1 <i>DePlot</i>	9
2.2.2 <i>Towards zero-shot understanding with large multimodal models (LMMs)</i>	10
2.3 SYNTHESISING INFORMATION AND RETRIEVAL AUGMENTED GENERATION (RAG).....	11
2.4 SUMMARY OF LITERATURE REVIEW.....	12
3. DETECTION PHASE	13
3.1 DATA COLLECTION.....	13
3.1.1 <i>IIIT-AR-13K Dataset</i>	13
3.1.2 <i>Proprietary Openspace Ventures Dataset (OSV Pitch Deck Dataset)</i>	13
3.2 DATA PROCESSING.....	14
3.2.1 <i>Processing IIIT-AR-13K</i>	14
3.2.2 <i>Processing OSV Pitch Deck Dataset</i>	14

3.2.3 Summary of Data Processing.....	15
3.3 DATA LABELLING.....	15
3.3.1 Definition of a Chart.....	15
3.3.2 Bounding Box Annotations.....	15
3.4 Data Splitting.....	17
3.4 CHART DETECTION MODEL TRAINING.....	18
3.4.1 YOLO Object Detection Models.....	18
3.5 EVALUATION AND COMPARISON OF CHART DETECTION MODELS.....	18
3.5.1 YOLOV8-Nano vs YOLOV8-Medium.....	18
3.5.2 YOLO vs Document Image Transformer (DIT).....	19
3.6 COMPLETION OF DETECTION PHASE.....	20
5. EXTRACTION PHASE	21
5.1 GROUNDED ZERO-SHOT UNDERSTANDING WITH LARGE MULTIMODAL MODELS (LMM) WITH IN-CONTEXT LEARNING.....	21
5.1.1 Adapting Set-of-Marks Prompting Strategy.....	22
5.1.2 Balancing Speed and Accuracy with Adapted Set-of-Marks.....	22
5.2 DATA STORAGE STRATEGY AND DATABASE DESIGN CONSIDERATIONS.....	23
5.2.1 PostgreSQL Schema Design.....	23
5.2.2 Vector Database Collection and Page-level Content as LlamaIndex Documents.....	24
5.3 INDEXING WITH VECTOR DATABASE.....	25
5.3.1 Chunking Strategy and Rationale.....	25
5.3.2 Embedding with Optimised Model Selection and Vector Database Loading.....	27
5.4 INTEGRATION WITH DETECTION PHASE THROUGH FASTAPI ROUTES.....	28
5.5 COMPLETION OF EXTRACTION PHASE.....	28
6. SYNTHESIS PHASE	29

6.2 USE CASE 1: SEMANTIC SEARCH FOR RELEVANT DOCUMENTS.....	31
6.2.1 FastAPI Route: Document Search and Interaction with Vector Database.....	31
6.2.2 Semantic Search Frontend Implementation.....	32
6.3 USE CASE 2: DOCUMENT SELECTION.....	32
6.3.1 FastAPI Route: Document Retrieval and Interaction with Relational Database.....	32
6.3.2 Document Selection Frontend Implementation.....	33
6.4 USE CASE 3: DOCUMENT UPLOAD.....	34
6.4.1 FastAPI Route: Document Upload.....	34
6.4.2 Document Upload Frontend Implementation.....	34
6.5 USE CASE 4: CHAT WITH DOCUMENT.....	35
6.5.1 FastAPI Routes: Document Information and PDF Rendering.....	36
6.5.2 FastAPI Route: Chat.....	36
6.5.3 Chat with Document Frontend Implementation.....	37
6.6 DEPLOYMENT ARCHITECTURE AND INFRASTRUCTURE.....	39
6.7 COMPLETION OF SYNTHESIS PHASE.....	40
7 EVALUATION OF FRAMEWORK	42
7.1 LIMITATIONS ON CURRENT BENCHMARKS.....	42
7.2 A HYBRID APPROACH TO BENCHMARKING.....	42
7.2.1 Evaluating Relevance with RAGAS.....	43
7.2.2 Evaluating Correctness through Surveys with Venture Capital Professionals.....	44
7.3 EVALUATION RESULTS.....	45
8. CONCLUSION	48
8.1. SUMMARY.....	48
8.2 LIMITATIONS.....	49

8.1.1 Reliance on Closed-source Models.....	49
8.1.2 Uncertainty in Model Outputs.....	50
8.1.3 Lack of a Rigorous Benchmark to Test Framework.....	50
8.3 RECOMMENDATIONS FOR FUTURE WORK.....	51
8.2.1 Controlling Model Deployments through Employment of Open-Source Models.....	51
8.2.2 Mitigating Uncertainty in Model Outputs.....	51
8.2.3 Creation of a Novel Benchmark Targeting Pitch Decks in a Deal-sourcing Context.....	53
REFERENCES	X
APPENDICES	XII
APPENDIX A: INTERVIEW SUMMARIES AND FINDINGS.....	XII
A.1 Summary of Interviews Conducted with Venture Capital Professionals.....	XII
APPENDIX B: DATA COLLECTION AND PROCESSING.....	XV
B.1 Data Processing Steps and Outcomes.....	XV
B.2 Details of the IIIT-AR-13K Dataset.....	XVI
APPENDIX C: CHART DETECTION MODEL TRAINING AND EVALUATION.....	XVII
C.1 Evaluation Metrics and Results.....	XVII
APPENDIX D: EXTRACTION PHASE DETAILS.....	XVIII
D.1 Adapted Set-of-Marks Strategy.....	XVIII
D.2 Data Storage and Database Design.....	XIX
APPENDIX E: EVALUATION FRAMEWORK AND METRICS.....	XX
E.1 RAGAS Context Relevancy Metric.....	XX
E.2 Metrics Related to Proposed Evaluation Framework.....	XX
E.3 Proposed Human Evaluation Framework (Survey).....	XXI
APPENDIX F: ADDITIONAL RESOURCES.....	XXI

F.1 Sample Pitch Deck Pages with Charts..... XXI

1. Introduction

1.1 Problem and Motivation

In the venture capital (VC) industry, data-driven decision-making has become increasingly crucial for identifying promising investment opportunities. Deal sourcing, which initiates the investment process, plays a pivotal role in driving the VC business.

However, this stage presents a significant bottleneck due to the reliance on manually analysing and synthesising information from visually rich, unstructured documents such as pitch decks and research reports.

These documents often contain critical information presented through various visual elements, including texts, tables, and charts. Venture capitalists must invest substantial time and effort to comprehend and extract insights from these visual representations, which can hinder the efficiency and effectiveness of the deal-sourcing process.

Through interviews with venture capital professionals (Appendix A), we identified that a significant portion of the information that drives decision-making in deal sourcing is derived from visually rich documents such as pitch decks and research reports. Notably, the charts present in these documents were highlighted as crucial sources of insights for evaluating potential investments.

1.1.1 An Emphasis on Pitch Deck Documents

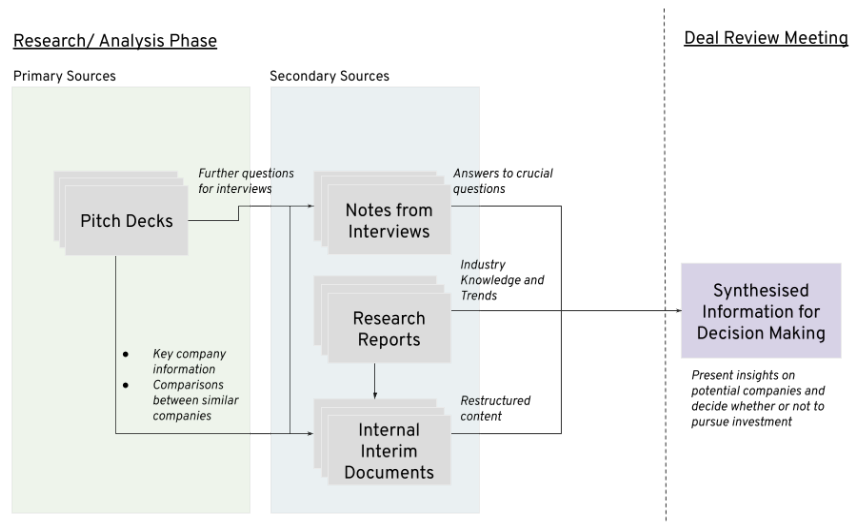


Figure 1: Diagram showing the information flow between documents during the deal-sourcing process in a VC firm

As shown in Figure 1, pitch decks serve as the initial point of contact between start-ups and venture capitalists in the deal-sourcing process. These documents are the primary means for start-ups to showcase their business potential and attract investment interest (Starttech, 2022). Given their critical role as the first touch point, pitch decks have a significant impact on the subsequent stages of the deal-sourcing process.

Thus, we focus our research on pitch decks, aiming to address the challenges associated with analysing and extracting insights from the charts contained within these documents. Developing a framework specifically tailored to pitch decks will enable venture capitalists to efficiently process and synthesise the visual information, leading to more informed and timely investment decisions.

The presence of charts in pitch decks shows the importance of visual representations in conveying complex information concisely and effectively. In the following section, we delve into the significance of charts in pitch decks and how they drive decision-making in the deal-sourcing process.

1.1.2 Importance of Charts in VC Deal Sourcing

Through interviews with venture capital professionals (Appendix A), we identified that a significant portion of the information that drives decision-making in deal sourcing is derived from visually rich documents such as pitch decks and research reports. Notably, the charts present in these documents were highlighted as crucial sources of insights in the evaluation of potential investments.

We assert that charts featured in pitch decks typically convey important information across the following aspects:

Aspect	Description	How it Drives Decision-making
Competitive Landscape	Charts often depict metrics related to the competitive landscape in which the start-up operates	Enables a comprehensive understanding of market dynamics and positioning of the company
Financial Metrics	Charts commonly present key financial metrics and performance indicators of the start-up, such as revenue growth, customer acquisition costs, and profitability projections	Critical for assessing the investment potential and financial viability of the venture

Table 1: Mapping of aspects communicated through charts and how they are pertinent in driving decision-making in deal sourcing in a VC context

1.1.3 Frequently Asked Questions (FAQs) in VC Deal Sourcing

Based on the identified aspects conveyed through charts in pitch decks, our interviews with venture capital professionals revealed a set of commonly asked questions that drive decision-making during the deal-sourcing process. These questions often revolve around understanding the competitive landscape and financial metrics of the start-up (Table 2)

S/N	Aspect	Subcategory	Frequently Asked Question (FAQ)
-----	--------	-------------	---------------------------------

1	Competitive Landscape	Competition	What is the start-up's competitive advantage/moat?
2		Market Size	What can we infer about the market size for the start-up's product?
3		Opportunities	What is the potential market opportunity for the start-up's product or service?
4	Financial Metrics	Current Traction	What is the start-up's revenue this year and last year, taking this year as 2024?
5			What are the start-up's expenses this year and last year, taking this year as 2024?
6		Unit Economics	What is the customer acquisition cost (CAC)?
7			What is the retention cost?
8			What is the customer lifetime value (CLTV)?
9		Profit and Loss (P&L) Shape	What are the profitability trends for the start-up?

Table 2: The FAQs in VC Deal Sourcing based on the aspects focused on charts

Accurately answering these frequently asked questions is crucial for venture capitalists to make informed investment decisions. However, manually extracting and synthesising the relevant information from visually rich documents, particularly charts, can be time-consuming and prone to errors.

To address this challenge, there is a pressing need for a data-driven framework that can efficiently extract and synthesise information from charts within unstructured documents, enabling targeted question-answering capabilities tailored to the VC deal-sourcing workflow.

1.2. Objectives and Solutions

To tackle the bottleneck in the deal-sourcing process and address the FAQs that can be answered through information found in pitch decks and charts identified in Section 1.1.2, we propose an end-to-end framework that comprises three distinct phases: Detection, Extraction, and Synthesis (Figure 2).

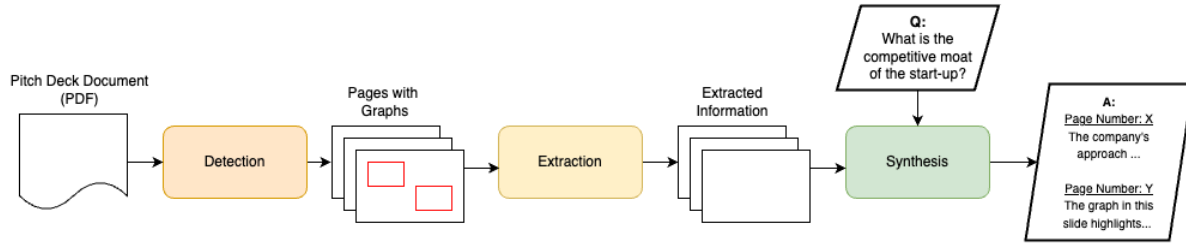


Figure 2: Overview of our framework to tackle bottlenecks in deal-sourcing: Detection, Extraction, Synthesis

The objectives of our framework are directly aligned with the user stories and requirements gathered through interviews with venture capital professionals (Table 3). These user stories highlight the need for a solution that enables venture capital analysts to quickly obtain answers to frequently asked questions related to the competitive landscape and financial metrics of a start-up and allows venture capital professionals to access synthesised information through an intuitive and user-friendly interface.

User Story	As a...	I want to...	So that...
1	Venture capital analyst	Quickly obtain answers to frequently asked questions (FAQs) related to the	I can efficiently evaluate investment opportunities

		competitive landscape and financial metrics of a start-up	and make informed decisions
2	Venture capital professional	Access the synthesised information through an intuitive and user-friendly interface	I can easily navigate through processed documents, ask questions, and obtain relevant insights

Table 3: User story table

To address these user stories, our framework employs techniques in document understanding, information extraction, and retrieval-augmented generation. The Detection Phase focuses on accurately identifying and localising chart instances within pitch decks. This phase leverages object detection techniques to locate chart instances, ensuring that subsequent phases have access to the necessary information for further analysis.

Building upon the output of the Detection Phase, the Extraction Phase aims to extract and understand the information conveyed by the detected chart instances. This phase employs multimodal models to comprehend the visual and textual content within the documents, storing the extracted information in a structured format for efficient retrieval and querying. Moreover, the Extraction Phase encompasses the understanding of user queries and intents, laying the foundation for the Synthesis Phase.

The Synthesis Phase is the culmination of our framework, directly addressing the user stories identified in Table 3. This phase focuses on synthesising the extracted information from each document to answer predefined questions crucial for due diligence and decision-making in the deal-sourcing context. By leveraging retrieval-augmented generation techniques, the Synthesis Phase provides the means to generate comprehensive responses to FAQs relevant to the key aspects identified in the charts, such as competitive landscape and financial metrics.

The synthesised insights can be efficiently accessed by venture capital professionals through multi-turn conversations presented in an intuitive, user-friendly interface. This aligns with User Story 2, empowering venture capital professionals to easily navigate through processed documents, ask questions, and obtain relevant insights. By streamlining access to critical

information, our framework aids in making informed investment decisions and addresses the bottlenecks in the deal-sourcing process.

2. Literature Review

The literature review explores three key areas relevant to our research: document image analysis, chart information extraction, and retrieval-augmented generation, each respectively related to the three-phase framework that we propose: Detection, Extraction, and Synthesis.

2.1 Document Image Transformer (DiT) and YOLO (You Only Look Once) in Document Image Analysis

Visual component extraction from unstructured documents is mostly approached as an object detection problem. Existing literature first converts document pages to images, then an object detection model is trained. In the research area of document understanding, this task is formalized as Document Layout Analysis - in that document semantics are understood by detecting instances of layout pieces, such as headings, titles, tables, and figures (Rastogi, 2023). Through the semantic understanding of unstructured documents, we hope to be able to enhance the extraction of information from charts in the later stages of the pipeline.

Li et al. propose a transformer-based model as the backbone, with Mask-RCNN as the detection head to perform Document Layout Analysis. Its specialized architecture comes from the use of a document-specific tokenizer the Masked Image Modelling step in pretraining - a tokenizer was trained on the IIT-CDIP dataset that includes 42 million document images (Rastogi, 2023).

However, we found that DiT + Mask-RCNN may not be very suitable for the use case, which prioritizes both speed and accuracy. The smallest DiT model, DiT-Base, already has 86M parameters. We expect that there may be latency issues when it comes to inference speed on the downstream. Furthermore, using DiT + Mask-RCNN off-the-shelf does not suffice in our use case. Purely using DiT + Mask-RCNN finetuned on the PubLayNet dataset has its associated issues. PubLayNet is a large-scale dataset comprising papers from PubMed, with 5 distinct class labels (Text, Title, List, Table, and Figure). Since we want to identify only chart instances, the ‘figure’ class of PubLayNet is too broad of a classification to encompass what we need.

Furthermore, the model, having trained on PubLayNet which is more geared towards medical research papers, will most likely suffer from poor performance on out-of-domain examples, like in our specific use case which focuses on pitch decks. Further fine-tuning of the model is likely needed.

Given the associated limitations of DiT for our use case, we approach Document Layout Analysis in a data-centric manner rather than model-centric. In other words, the running hypothesis is that smaller models, having sufficiently trained on examples catered to the deal-sourcing context, can outperform the larger Document Image Transformer model that is more highly specialized in Document Layout Analysis.

The advent of You Only Look Once (YOLO) models originated before the transformers, and subsequently, vision transformers and DiT came into the picture. The original YOLO paper proposed an efficient method to approach object detection - that is reframing it as a single-pass regression problem. Compared to its two-stage detector predecessors which split detection into regression and classification, YOLO excels in its lightweight approach (Hussain, 2023).

The latest iteration of YOLO is YOLOv8 by Ultralytics. Although architectural specifics of YOLOv8 have not been released, benchmark studies show that YOLOv8 outperforms all its predecessors, achieving lower latency and higher accuracy with similar parameters (Hussain, 2023). Given its strength in achieving high accuracy and low latency, our study experiments with YOLO models such as YOLOv8-Nano (3M parameters) and YOLOv8-Medium (25M parameters), then comparing their performance against DiT in terms of speed. The metric that shall be used will be speed in milliseconds (ms) inference per image.

2.2 ChartQA and Extracting Information from Charts

Most of the existing literature centres on images of charts as inputs, and outputs being text or tables. Such literature is centred on the use of transformer-based models to accomplish the said tasks. The goals of such conversions, though, slightly differ from each other.

Notable datasets and benchmarks that go about validating the task of information extraction from charts include PlotQA, ChartQA, and Chat-to-text.

PlotQA (Methani et al., 2020) and ChartQA (Masry et al., 2022) are similar in that the input would be a question, and the output would be an answer to that question, contextualized on information extracted from charts. Chart-to-text (Obeid & Hoque, 2020) also focuses on information extraction from charts, but goes about it in a slightly different way - in that chart images are converted to natural language summaries.

The task of question-answering over charts is achieved by converting chart images into natural language descriptions, which is the task of chart-to-text. A large language model is then augmented with these natural language descriptions and finally used to answer a given query. Thus, the task of chart understanding can be generalized to converting chart images into natural language.

2.2.1 DePlot

The current state-of-the-art is DePlot (Liu et al., 2023). They standardize the chart image-to-natural language tasks by proposing to convert the input chart image to a linearised table and passing the linearised table to LLMs for one-shot reasoning. The training is done by performing finetuning on a previous checkpoint of a transformer-based model, MatCha (Liu et al., 2023), an image-encoder-text-decoder. As for the datasets involved, DePlot was mostly trained with generated synthetic data and crawled real-world data, specifically from four websites - [statista.com](https://www.statista.com), [pewresearch.com](https://www.pewresearch.com), [ourworldindata.org](https://www.ourworldindata.org), and [oecd.org](https://www.oecd.org).

A drawback of DePlot lies in the datasets involved during training. A commonality in generated synthetic data and the data from the four websites mentioned above is the need for more diversity of charts. Generated synthetic charts follow common styles through plotting libraries such as `matplotlib` and `seaborn`. Though data from the four websites offer greater diversity, they are not diverse enough for our context, where charts inherently do not adhere to common styles and conventions. For example, some charts in the OSV pitch deck dataset are bar charts without clear axes (Appendix B Figure 20 & Figure 23); line charts with no point annotations (Appendix B Figure 22); and pie charts with no labels (Appendix B Figure 21). This means that finetuning is likely needed.

The next drawback of DePlot lies in the high cost of annotations. As fine-tuning is needed, this means that plot-table pairs need to be obtained from the OSV pitch deck dataset. Tables

annotations are expected to be highly costly in the long term, thus the use of DePlot is not a viable option in chart understanding.

2.2.2 Towards zero-shot understanding with large multimodal models (LMMs)

Set-of-Mark prompting proposed by Yang et al. shows that marked input images strengthen the reasoning capabilities of LMMs through visual grounding. Thus, we propose to adapt the Set-of-Mark prompting strategy in document images, with the marks specifically being the bounding box predictions of chart instances obtained from Document Image Analysis.

Our proposed strategy has pros aside from the general strengthened reasoning capabilities brought about by Set-of-Marks. First, inputting the entire document page image allows the LMM to associate the charts with other regions of the page. This allows for a more comprehensive analysis and extraction of information that can answer the questions for our use case.

We estimate that the output from this strategy would be significantly better than the supervised fine-tuning approach with DePlot, which takes in only the chart instances as input. Second, sending only marked images as input reduces the time needed to process a question, as opposed to sending all document images as input to the LMM.

However, there are cons associated with our proposed strategy as well. It will be difficult to evaluate the output of our strategy. Though we can use benchmarks such as ChartQA (Masry et al., 2022), PlotQA (Methani et al., 2020), and chart-to-text, the data is not representative of what we have in the OSV pitch deck dataset, as mentioned in the drawbacks of DePlot.

2.3 Synthesising Information and Retrieval Augmented Generation (RAG)

Retrieval Augmented Generation (RAG) is a promising approach for synthesising information extracted from charts. RAG enhances language models by integrating external knowledge retrieval into the generation process, addressing limitations such as hallucinations (Lewis et al., 2020).

The RAG framework has evolved from Naive RAG, which follows a traditional process of indexing, retrieval, and generation, to Advanced RAG, which introduces optimizations in data indexing, retrieval, and post-retrieval processing to suit specific contexts (Wang et al., 2022).

We will focus on optimizing the different stages of RAG: indexing, retrieval, generation, and augmentation. In the indexing and retrieval stages, careful consideration will be given to data chunking and granularity of representation to effectively retrieve and synthesise information from the extracted charts. Chunking entails that text is divided into smaller chunks, which are then encoded using an embedding model and stored in a vector database. This not only helps overcome the context limitations of language models (Wang et al., 2022) but helps the subsequent retrieval stage, in which similarity searches are performed on the embedded chunks.

Next, during the generation stage, post-retrieval processing methods will be employed to refine the retrieved information and align it with the specific requirements of our task. Finally, in the augmentation stage, we will effectively combine the retrieved information with the original query to generate informative and contextually relevant responses.

By leveraging the advancements in RAG and tailoring the approach to our specific use case, we aim to develop an effective solution for synthesising information from the extracted charts, ultimately enabling venture capitalists to make data-driven investment decisions.

2.4 Summary of Literature Review

Research Area	Current Approaches	Limitations	Our Proposed Solution	Mapping to Phase
Document Image Analysis	Document Image Transformer (DiT) YOLO (You Only Look Once)	DiT has a large parameter count and potential performance issues on out-of-domain examples. DiT is also less suitable for use	A data-centric approach using smaller models like YOLO, training on examples specific to the	Detection Phase

		cases prioritising speed and accuracy	deal-sourcing context	
Chart Information Extraction	Datasets and benchmarks: PlotQA, ChartQA, Chart-to-text State-of-the-art-model : DePlot	DePlot’s data lack of diversity, and it is costly to obtain annotations required for fine-tuning DePlot	Adapting Set-of-Marks prompting strategy with LMMs, enabling zero-shot understanding of charts in the context of the entire document page	Extraction Phase
Retrieval Augmented Generation (RAG)	RAG enhances language models by integrating external knowledge retrieval	RAG when performed naively can yield poor results	Optimising indexing, retrieval, and generation stages of RAG, tailoring the approach to the specific use case in the VC domain	Synthesis Phase

Table 4: Summary of Literature Review

3. Detection Phase

The Detection Phase is the first stage in our end-to-end framework, focusing on accurately identifying and localizing chart instances within unstructured venture capital documents, specifically pitch decks. This phase involves several key steps: data collection, processing, labelling, and model training. By leveraging state-of-the-art object detection architectures and training them on our domain-specific datasets, we aim to develop robust models capable of accurately detecting charts in visually rich documents.

3.1 Data Collection

3.1.1 IIIT-AR-13K Dataset

The IIIT-AR-13K dataset (Mondal et al., 2020) is a document image analysis dataset, specifically targeting annual reports. Five categories - ‘table’, ‘figure’, ‘natural image’, ‘logo’, and

‘signature’ are manually annotated in publicly available annual reports. In 4.2 Data Processing, we discuss how we narrow the classes down to the target class that we need for our use case, which is the ‘chart’ class.

The IIIT-AR-13K Dataset is chosen because annual reports and pitch decks share similarities in terms of containing finance-related charts. These charts are identified as crucial for answering key questions that drive decision-making in venture capital, as introduced in section 1.1.3.

3.1.2 Proprietary Openspace Ventures Dataset (OSV Pitch Deck Dataset)

Openspace Ventures has provided us with access to a collection of 20,052 documents, including pitch decks, financial reports, and operational documents.

To curate a dataset specific to our use case, we define heuristics (Appendix B, Figure B.1) to filter out potential pitch decks from the collection. This initial filtering process yielded 928 candidate files, which were further refined through manual inspection, resulting in a final set of 874 pitch deck documents.

We converted each pitch deck into a series of images representing each document page, which were then annotated using the Roboflow platform (Roboflow, 2024). To the best of our knowledge, the OSV Pitch Deck Dataset is a novel dataset focused specifically on pitch deck documents, offering a diverse range of chart styles and designs. This diversity in chart representations is crucial in advancing the field of visually rich document understanding, particularly in the context of chart analysis.

3.2 Data Processing

3.2.1 Processing IIIT-AR-13K

From the IIIT-AR-13K dataset, which contains 13,408 images (train, validation, test combined), we narrowed our focus to the ‘figure’ class, as it encompasses chart instances. Initially, 1,849 images were identified as containing ‘figure’ instances (Appendix B, Figure B.2.1). However, upon further manual inspection, we discovered that not all ‘figure’ instances matched our criteria for charts Appendix B. Figure B.2.2. Consequently, we performed additional processing on the

Roboflow platform, resulting in 1,352 images containing only chart instances relevant to our use case.

3.2.2 Processing OSV Pitch Deck Dataset

The proprietary OSV Pitch Deck Dataset, comprising 874 documents, underwent a rigorous manual processing phase to remove documents without any chart instances. This step left us with 686 relevant documents. We then converted each document into a series of .JPG format images using the pdf2image library (Belval, n.d.), with fixed dots per inch of 300, a measure of image resolution. From these images, we identified 2,348 instances containing charts, which were subsequently annotated with bounding boxes on the Roboflow platform.

3.2.3 Summary of Data Processing

The data processing step yielded the following breakdown of images across the two datasets (Table 5):

Dataset	Number of Images		
	Original	With Figure Instances	With Chart Instances
IIIT-AR-13K	13,408	1849	1352
OSV Pitch Deck Dataset	21,229 (874 Documents)	-	2348 (686 Documents)

Table 5: Summary of Data Processing

3.3 Data Labelling

3.3.1 Definition of a Chart

For purposes of data labelling, we define a chart as:

"Any semblance to a pie chart, line chart, or bar chart is considered a chart, regardless of whether their axes are present. Specifically, a pie chart is any circular shape representing a proportional breakdown between different categories; a bar chart is any bar or set of bars representing a quantity or quantities of different categories; and a line chart is any line or set of lines representing a quantity changing over a dimension, e.g., time."

While our definition focuses on pie, bar, and line charts, we intentionally kept it broad to account for the visually rich and diverse nature of charts present in the OSV Pitch Deck Dataset. (Appendix F.1).

3.3.2 Bounding Box Annotations

Using the Roboflow platform, we manually annotated the OSV Pitch Deck Dataset with bounding boxes based on our definition of a chart. For the IIIT-AR-13K Dataset, we removed annotations that did not match our criteria, resulting in an updated dataset called IIIT-AR-13K-Charts. The annotations were then exported to the YOLOV8 format to facilitate model training.

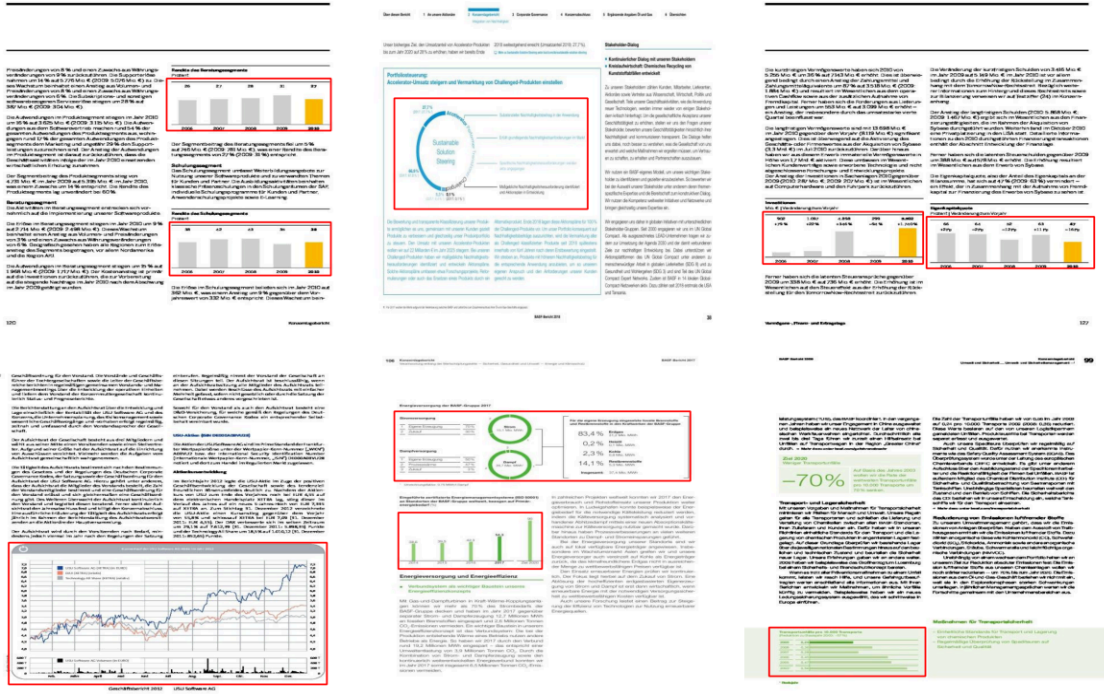


Figure 3: Sample of chart annotations on images from IIIT-AR-13K-Charts

<removed>

Figure 4: Sample of chart annotations on images from OSV Pitch Deck Dataset

3.4 Data Splitting

To train our object detection models, we adopted a document-wise train-test-split methodology, as proposed by Mondal et al. This approach involves splitting the dataset at the document level rather than treating all document images independently, mitigating the risk of data leakage.

For the IIIT-AR-13K dataset, we allocated the 1,352 images with chart instances back into their original train-validation-test splits. As for the OSV Pitch Deck Dataset, we split the 686 documents into train-validation-test splits with an approximate 80:10:10 ratio.

Subsequently, we combined the IIIT-AR-13K-Charts and OSV Pitch Deck Dataset to form our final dataset for object detection model training, which we refer to as the IIIT-OSV-Charts dataset. The final dataset specifications are as follows:

IIIT-OSV-Charts Dataset	Train	Validation	Test
# Images from IIIT-AR-13K	1035 (76.6%)	61 (4.5%)	256 (18.9%)
# Images from OSV Pitch Deck Dataset (686 Documents)	1902 Images (81.01%)	172 Images (7.33%)	274 Images (11.67%)
Total # Images	2937 (79.38%)	233 Images (6.30%)	530 Images (14.32%)

Table 6: Final Dataset Specifications

This rigorous data processing, labelling, and splitting approach ensured that our final dataset accurately represented the diverse chart representations found in pitch decks while adhering to best practices to mitigate potential issues such as data leakage.

3.4 Chart Detection Model Training

3.4.1 YOLO Object Detection Models

Moving beyond the theoretical comparison of model architectures in the literature review, we validated our hypothesis by building the best chart detection model for our use case. The primary objective was to achieve both speed and accuracy in chart detection.

We experimented with two variants of the YOLO (You Only Look Once) object detection models: YOLOV8-Nano and YOLOV8-Medium. YOLOV8-Nano is a lightweight model with only 3 million parameters, making it suitable for resource-constrained environments. In contrast, YOLOV8-Medium is a larger model with 25 million parameters, offering a balance between speed and accuracy. By evaluating the performance of these models on our domain-specific IIIT-OSV-Charts dataset, we aimed to identify the most appropriate architecture for chart detection in the deal-sourcing context.

We trained both YOLOV8-Nano and YOLOV8-Medium on the curated IIIT-OSV-Charts dataset, logging all results on the Weights and Biases platform (Weights & Biases, 2024). Over 100

epochs, both models exhibited similar patterns in terms of evaluation metrics such as precision, recall, mAP50, and mAP50-95. All training was performed on Paperspace cloud-compute instances, specifically using the P5000 with 30GiB RAM, 8 CPUs, and 16 GiB GPU (Paperspace, 2024).

The training and validation losses followed similar patterns (Appendix C.1), indicating that a higher number of parameters did not necessarily translate to better model performance. In this case, the larger YOLOV8-Medium model did not outperform the smaller YOLOV8-Nano model.

3.5 Evaluation and Comparison of Chart Detection Models

3.5.1 YOLOV8-Nano vs YOLOV8-Medium

The evaluation metrics on the test set are shown as such:

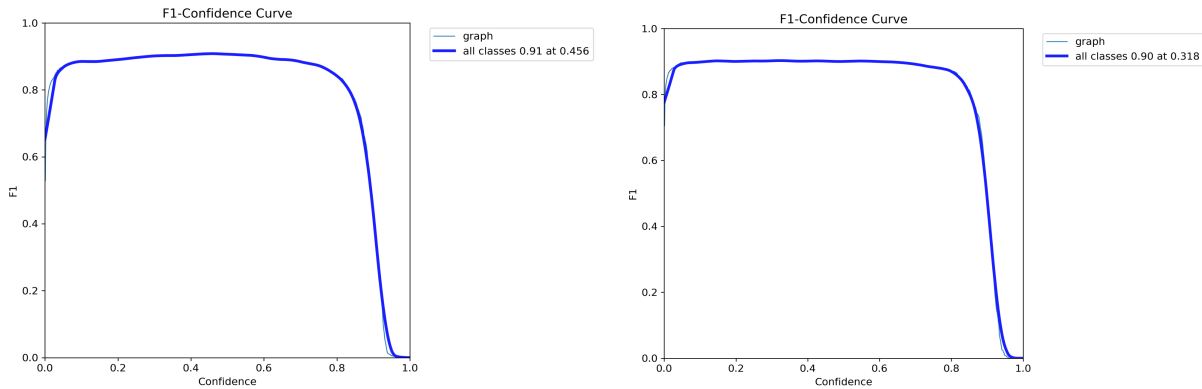


Figure 5: F1-Confidence Curve of YOLOV8-Nano (Left) and YOLOV8-Medium (Right) on IIT-OSV-Charts test set.

From Figure 5, both YOLO models exhibited similar performance, despite differences in the number of parameters. YOLOV8-Nano achieved a maximum F1 of 0.91 at the threshold of 0.456, while YOLOV8-Medium achieved a maximum F1 of 0.90 at the threshold of 0.318.

Even with the use of a small model optimised for faster inference speeds, we achieved comparable results to the larger model. Thus, we proceeded with the fine-tuned YOLOV8-Nano model during this research. YOLOV8-Nano achieves an average speed of 2.8ms per inference, significantly faster than YOLOV8-Medium’s average speed of 12.9ms per inference.

3.5.2 YOLO vs Document Image Transformer (DIT)

We also conducted experiments with the Document Image Transformer (DIT), specifically the DIT-Base model fine-tuned on the PubLayNet dataset. Although we did not fine-tune DIT-Base on the IIIT-OSV-Charts dataset due to latency concerns, we evaluated its performance on a sample of documents from the OSV dataset. The average inference time for DIT-Base was 5.9 seconds, significantly slower than the trained YOLO models. These results validated our hypothesis that smaller models, when sufficiently trained on examples tailored to the deal-sourcing context, can outperform larger, specialised models like DIT in Document Layout Analysis. Thus, we proceeded with the fine-tuned YOLOV8-Nano model for the remainder of the research.

	YOLOV8-Nano fine-tuned with IIIT-OSV-Charts Dataset	YOLOV8-Medium fine-tuned with IIIT-OSV-Charts Dataset	DiT-Base fine-tuned on PubLayNet
Number of Parameters	3M	25M	87M
Speed (ms)	2.8	12.9	5900

Table 7: Comparison of YOLO models and DIT in the number of parameters and inference speed

3.6 Completion of Detection Phase

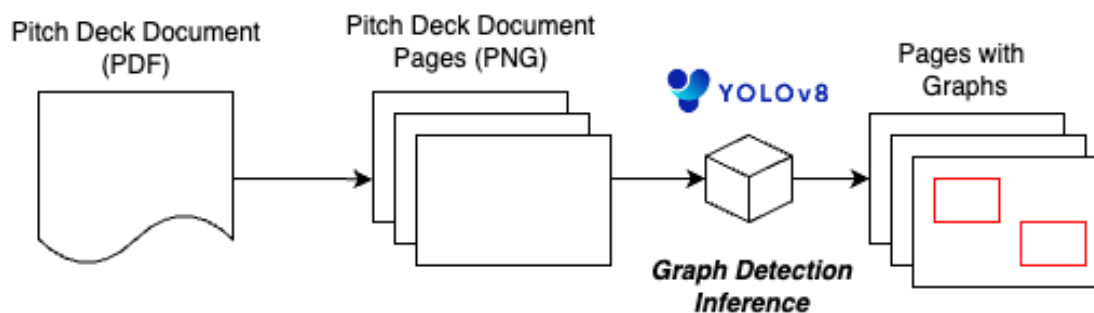


Figure 6: Summary of Detection Phase

In the Detection Phase, we made significant contributions by creating a novel dataset, the IIIT-OSV-Charts dataset, which combines the IIIT-AR-13K dataset and the proprietary OSV Pitch Deck Dataset. This dataset is specifically curated for training chart detection models in the context of venture capital documents, addressing the lack of publicly available datasets in this domain.

Through our experiments, we validated the hypothesis that smaller models, when sufficiently trained on examples catered to the deal-sourcing context, can outperform larger, specialized models like the Document Image Transformer (DIT). We found that the fine-tuned YOLOV8-Nano model achieved a maximum F1 score of 0.91 on the IIIT-OSV-Charts test set while maintaining an average inference speed of 2.8ms per image. This demonstrates the effectiveness of our data-centric approach in developing a fast and accurate chart detection model.

To facilitate seamless integration with the subsequent Extraction Phase, we saved the fine-tuned YOLOV8-Nano model as a TorchScript. This allows for efficient inference and enables the chart detection functionality to be easily incorporated into the overall framework.

With the completion of the Detection Phase, we have laid a strong foundation for accurately identifying and localizing chart instances within pitch decks. The output of this phase, in the form of bounding box coordinates for each detected chart, will serve as the input for the Extraction Phase, where we will focus on extracting and understanding the information conveyed by the identified charts.

5. Extraction Phase

The Extraction Phase focuses on extracting and understanding the information conveyed by the detected chart instances from the previous Detection Phase. This phase involves several key steps: grounded zero-shot understanding using large multimodal models (LMMs), storing the extracted information in a vector database, and implementing the necessary components for efficient extraction and retrieval. By leveraging the power of LMMs and the Set-of-Marks prompting strategy, we aim to extract insights from the detected charts and their surrounding context.

5.1 Grounded Zero-shot Understanding with Large Multimodal Models (LMM) with In-context Learning

To extract insights from the detected charts, we employ a grounded zero-shot understanding approach using Large Multimodal Models (LMM). This approach allows us to leverage the pre-trained knowledge of LLMs while providing them with relevant context to guide the extraction process.

5.1.1 Adapting Set-of-Marks Prompting Strategy

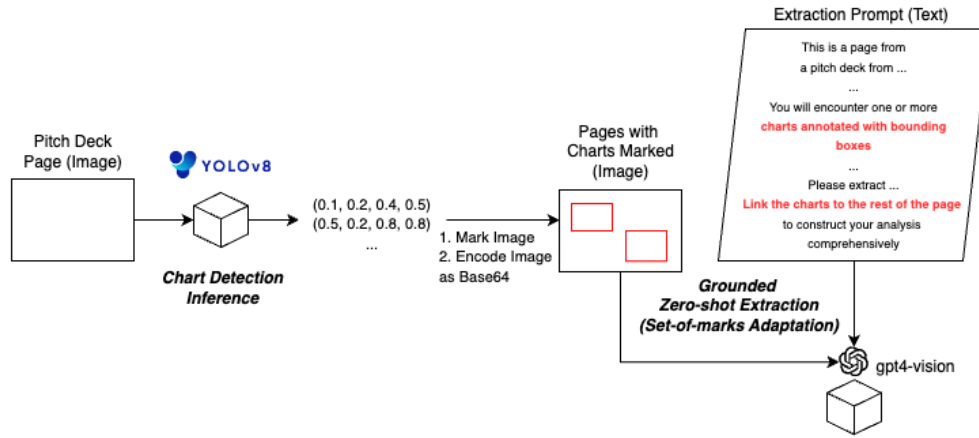


Figure 7: Our adapted Set-of-Marks prompting strategy

We adapt the Set-of-Marks prompting strategy, proposed by Yang et al., to specifically ground the LMM on document pages containing chart instances. Unlike the original open-vocabulary-based grounding, our targeted approach leverages the bounding box coordinates obtained from the supervised chart detection model. By associating the chart instances strongly with a carefully crafted prompt (Appendix D, Listing D.1), we further strengthen the LMM's understanding through in-context learning.

Our adaptation involves sending the entire document page image containing chart instances as input to the LMM, along with the corresponding bounding box coordinates (Figure 7). This approach enables the LMM to associate the charts with other relevant regions on the page,

facilitating a more comprehensive analysis and extraction of information from the visual and textual context.

5.1.2 Balancing Speed and Accuracy with Adapted Set-of-Marks

In our approach to extracting information from detected chart instances, we paid careful attention to balancing speed and accuracy. While leveraging LMMs and the adapted Set-of-Marks prompting strategy, we implemented a strategic processing approach to optimize both aspects.

Instead of indiscriminately sending all document pages to the LMM for processing, we selectively focused on the pages containing chart instances. This decision was driven by our understanding that charts serve as critical anchors for grounding the LMM's understanding, providing the necessary context to effectively answer the frequently asked questions (FAQs) in the venture capital domain.

By concentrating our efforts on the most relevant visual information – the pages with charts – we aligned our approach with the findings of Yang et al., who demonstrated that marked images strengthen the reasoning capabilities of LMMs through visual grounding. This targeted processing not only reduced computational overhead but also minimized the processing time by avoiding the analysis of irrelevant or redundant pages.

Furthermore, we employed asynchronous programming techniques using the `asyncio` library to further optimize for speed. Our implementation performs multiple concurrent requests to the LMM API (OpenAI GPT4-Vision), enabling the simultaneous processing of multiple pages. This concurrent execution significantly reduced the overall latency, contributing to a more efficient extraction process.

5.2 Data Storage Strategy and Database Design Considerations

After extracting information using our adapted Set-of-Marks approach, we implemented a structured data storage strategy to facilitate efficient retrieval and querying in subsequent phases. This involved carefully designing and leveraging two different database systems: a relational database and a vector database.

5.2.1 PostgreSQL Schema Design

We employed PostgreSQL (PostgreSQL, 2024), a relational database management system, to store metadata related to the detected chart instances and their corresponding documents. The design of our PostgreSQL schema was primarily driven by the requirements of the web application, which we would use to present synthesised information in the Synthesis Phase.

Since the primary objective of the web application is to enable direct querying of individual documents to answer frequently asked questions (FAQs), we structured our relational database to support efficient retrieval of document metadata, such as chart bounding boxes and the pages on which these bounding boxes are located. This design choice facilitated seamless integration with the subsequent Synthesis Phase, where we would leverage this metadata to provide context-aware responses tailored to each document. It is worth noting that we stored the bounding box-related data at this stage, laying the foundation for future phases (Appendix D, Figure D.2).

5.2.2 Vector Database Collection and Page-level Content as LlamaIndex Documents

To perform Retrieval-Augmented Generation (RAG), a critical component of our framework, we prepared our extracted data for storage in a vector database. We chose Qdrant (Qdrant, 2024), a scalable and efficient vector database solution, and interfaced with it using LlamaIndex (LlamaIndex, 2024), a modular library for text processing, indexing, and retrieval from various vector databases.

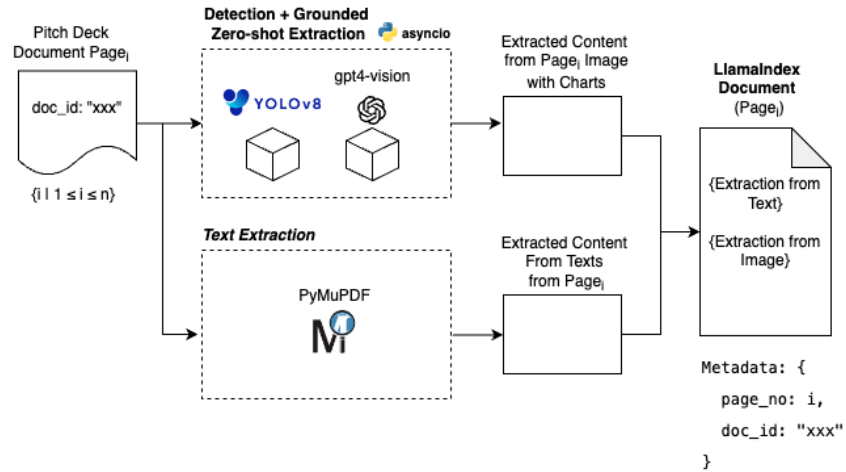


Figure 8: How page data is stored as LlamaIndex Documents

Based on interviews conducted with venture capital professionals (Appendix A), we found that they often analyse pitch decks on a page-by-page basis. Each page in a pitch deck tends to be crafted to convey specific information, such as financial projections, market opportunities, or product features. To align our approach with the natural way in which pitch decks are consumed and analysed, we opt to store data on a pitch deck page level.

We leveraged the LlamaIndex Document class, a generic container, to store the extracted content from each pitch deck page. Specifically, we combined the output from our grounded zero-shot understanding approach with the text extracted directly from the page using PyMuPDF (Pymupdf, 2024). This consolidated content was then stored as an instance of a LlamaIndex Document, with the page number and a unique pitch deck document ID included as metadata (Figure 8). This structured approach laid a solid foundation for efficient retrieval and generation in the subsequent Synthesis Phase.

5.3 Indexing with Vector Database

Having stored the page-level data in LlamaIndex Documents, we proceeded to obtain vector representations of the pitch deck documents. This indexing process is crucial as it sets the stage for efficient retrieval and generation in the subsequent Synthesis Phase.

5.3.1 Chunking Strategy and Rationale

Chunking is a key step in the indexing process, where text is divided into smaller units and encoded into vector representations using an embedding model. In the context of LlamaIndex, LlamaIndex Documents are typically further chunked into LlamaIndex Nodes, which serve as the atomic representations of data. Common chunking strategies involve dividing texts into sentences or paragraphs, with optional overlaps between chunks to preserve context (LlamaIndex, 2024).

In our approach, we carefully evaluated the necessity of further chunking our LlamaIndex Documents, which already represented individual pitch deck pages. Our goal was to align with the natural workflow of venture capital professionals, who typically analyse pitch decks on a page-by-page basis to obtain answers to frequently asked questions (FAQs) within the context of each page.

Hence, we proposed to treat each document page as a single chunk, eliminating the need for further chunking from LlamaIndex Documents to LlamaIndex Nodes (Figure 9). This approach balances between preserving the contextual integrity of the information and enabling efficient embedding and retrieval operations.

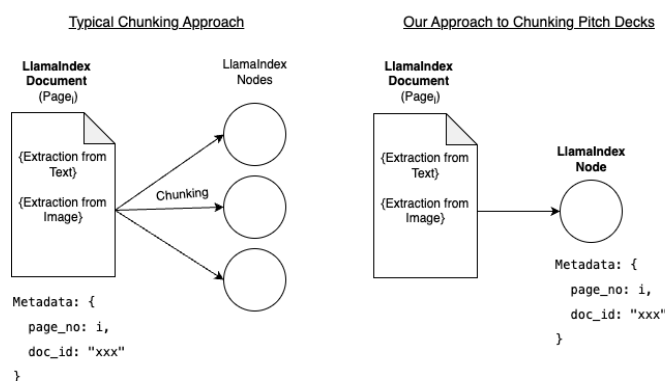


Figure 9: Typical approaches of obtaining LlamaIndex Nodes (left) vs. our approach for our use case (right)

By treating a document page as a single chunk, we hypothesized that pitch deck pages are generally not excessively wordy and can be effectively embedded without the need for further

chunking. To validate this hypothesis, we performed a distribution analysis on a sample of 100 documents from the OSV pitch deck dataset.

We measured the token count for each page's content using the tiktoken tokenizer, which was also used to tokenize inputs for the OpenAI text-embeddings-small embedding model. The analysis revealed that the distribution of tokens across pitch deck pages exhibited a right-skewed behaviour, with the maximum token count well below the 8,191 token limit of the text-embeddings-small model (OpenAI, 2024) (Figure 10).

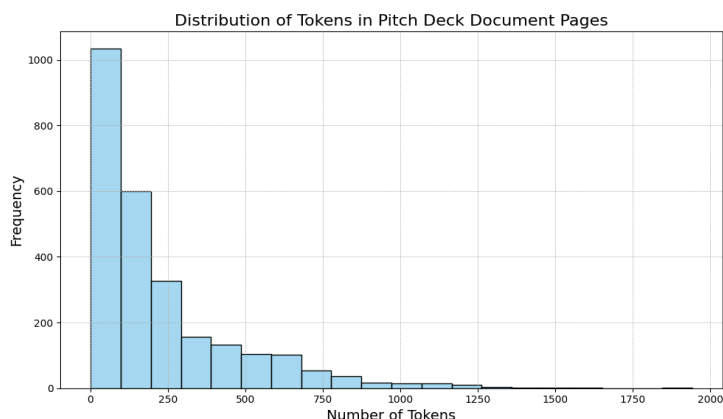


Figure 10: Distribution of tokens in pitch deck document pages from the OSV Pitch Decks Datasets

Based on these findings, we could infer that pitch deck pages are not overly wordy and that they can be embedded without further chunking. Although there is a risk of distribution shifts over time, we assert that this risk is negligible. This is because pitch decks are typically designed to be concise and visually engaging, focusing on delivering key information (Starttech, 2024). This corroborates our hypothesis and supports our decision to treat each page as a single chunk in our indexing stage, aligning with the natural analysis workflow of venture capital professionals.

5.3.2 Embedding with Optimised Model Selection and Vector Database Loading

After establishing our chunking strategy, we employ the OpenAI text-embeddings-small model to convert the chunks into vector representations. The choice of this model is based on careful considerations of indexing and retrieval speed, as well as retrieval quality.

While larger embedding models like `text-embeddings-large` offer higher dimensionality (3072 dimensions), they come with increased computational complexity and memory requirements. In contrast, `text-embeddings-small` provides a balanced trade-off between efficiency and effectiveness, with a dimensionality of 1536. This smaller model significantly reduces the memory footprint and enables faster vector searches and insertion times compared to its larger counterpart (OpenAI, 2024).

Moreover, extensive evaluations using the MTEB benchmark have demonstrated that the performance of `text-embeddings-small` is not significantly inferior to `text-embeddings-large` across a wide range of natural language understanding tasks (OpenAI, 2024). This indicates that the smaller model can maintain a high level of retrieval quality while offering substantial speed advantages. Given our framework's emphasis on fast retrieval and efficient memory utilization, OpenAI's `text-embeddings-small` emerges as the optimal choice.

Once the vector representations are obtained, we load them into a Qdrant database, organizing all pitch deck documents under a single Qdrant collection. This unified storage approach lays the foundation for semantic search capabilities in the upcoming Synthesis Phase. By consolidating the vector representations within a single collection, we can perform efficient and accurate similarity searches across the entire corpus of pitch decks.

5.4 Integration with Detection Phase through FastAPI Routes

In the development of our framework, we employed FastAPI as the backend, enabling seamless integration between the Detection and Extraction Phases. The `/upload` route facilitates the transfer of the output from the Detection Phase to the Extraction Phase for further processing.

In the `/upload` route, the `DocUploader` class is instantiated, and the `parse` method is called. We developed a `DocUploader` class to encapsulate the core functionality required to handle uploaded documents and extract relevant information from them. It first reads the uploaded document, performs pre-processing steps, and applies the chart detection model from the Detection Phase to identify chart instances within the pitch deck document. The class then passes the pages with charts to our adapted set-of-marks strategy for zero-shot understanding, combines output with

extracted text, performs post-processing in the form of conversions to data storage formats in 5.2, then loads to the PostgreSQL database and Qdrant database respectively (Figure 11).

5.5 Completion of Extraction Phase

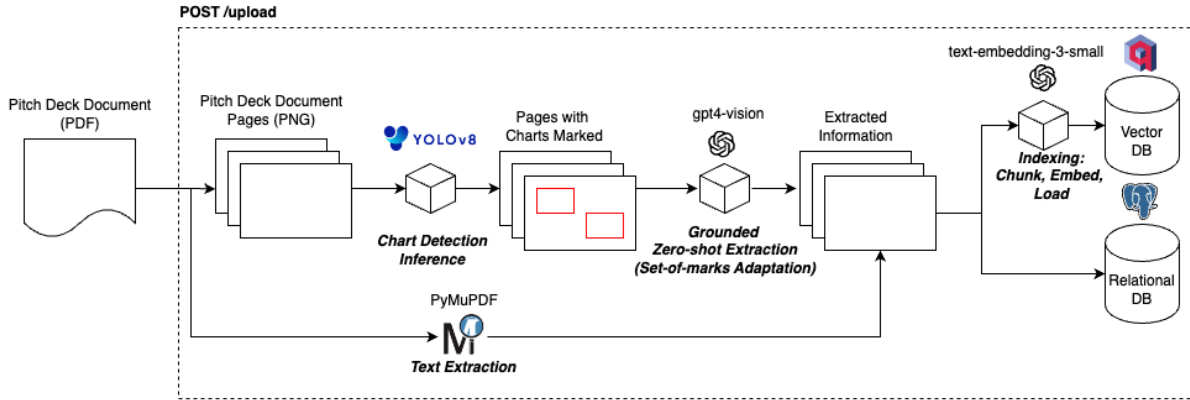


Figure 11: Summary of the Extraction Phase

In the Extraction Phase, we successfully integrated and built upon the outputs from the Detection Phase, effectively extracting and understanding the information conveyed by the identified chart instances within pitch deck documents. By leveraging large multimodal models (LMMs) in conjunction with our adapted Set-of-Marks prompting strategy, we achieved grounded zero-shot understanding of the charts within the context of the entire document page (Figure 11).

Our approach strikes a balance between speed and extraction quality. By selectively processing only the pages containing chart instances, we reduced computational overhead while maintaining the relevance of the extracted information. The strategic use of asynchronous libraries further optimized the extraction process, enabling concurrent processing of multiple pages and reducing overall latency.

To ensure efficient storage and retrieval of the extracted information, we designed a structured data storage strategy. We employed PostgreSQL for storing document metadata and the vector database Qdrant for storing page-level content. The integration of LlamaIndex provided a modular approach to indexing and querying the extracted information, laying the groundwork for the subsequent Synthesis Phase.

Furthermore, the development of the /upload FastAPI route played a pivotal role in bridging the Detection and Extraction Phases. This route facilitated the seamless transition between the two phases, enabling the extracted information to be passed on to the final phase, the Synthesis Phase, where it will be synthesised to generate comprehensive responses to frequently asked questions (FAQs) in the deal-sourcing context (Figure 11).

6. Synthesis Phase

The Synthesis Phase marks the culmination of our end-to-end framework, building upon the accomplishments of the Detection and Extraction Phases to deliver a user-centric solution that addresses the challenges faced by venture capital professionals in the deal-sourcing process. As highlighted in the Introduction (Section 1), the manual analysis of visually rich documents, such as pitch decks, is a time-consuming task that hinders the efficiency and effectiveness of decision-making in venture capital.

To tackle these challenges and align with the objectives outlined in Section 1.2, the Synthesis Phase focuses on synthesising the extracted information and presenting it through an intuitive web application. By empowering venture capital professionals with efficient access to relevant insights, we aim to streamline their decision-making process and enable them to make informed investment decisions.

The Synthesis Phase directly addresses the user stories identified in Section 1.2 (Table 3), providing venture capital analysts with the ability to quickly obtain answers to frequently asked questions (FAQs) related to the competitive landscape and financial metrics of a start-up. This capability enables them to efficiently evaluate investment opportunities. Furthermore, it offers venture capital professionals an intuitive and user-friendly interface to access synthesised information, allowing them to easily navigate through processed documents, ask questions, and obtain relevant insights. To achieve these objectives, the Synthesis Phase encompasses four distinct use cases. These use cases are based on our proposed user flow (Figure 12), which aligns with the identified user stories and objectives (Table 8).

Objective	User Story	Related Use Cases
-----------	------------	-------------------

Enable venture capital analysts to quickly obtain answers to frequently asked questions (FAQs) related to the competitive landscape and financial metrics of a start-up, allowing them to efficiently evaluate investment opportunities and make informed decisions	As a venture capital analyst, I want to quickly obtain answers to frequently asked questions (FAQs) related to the competitive landscape and financial metrics of a start-up, so that I can efficiently evaluate investment opportunities and make informed decisions	• Use Case 1: Semantic Search for Relevant Documents • Use Case 4: Chat with Document
Provide venture capital professionals with an intuitive and user-friendly interface to access synthesised information, enabling them to navigate through processed documents, ask questions, and obtain relevant insights	As a venture capital professional, I want to access the synthesised information through an intuitive and user-friendly interface, so that I can easily navigate through processed documents, ask questions, and obtain relevant insights	• Use Case 2: Document Selection • Use Case 3: Document Upload • Use Case 4: Chat with Document

Table 8: Mapping between objectives, user stories, and related use cases

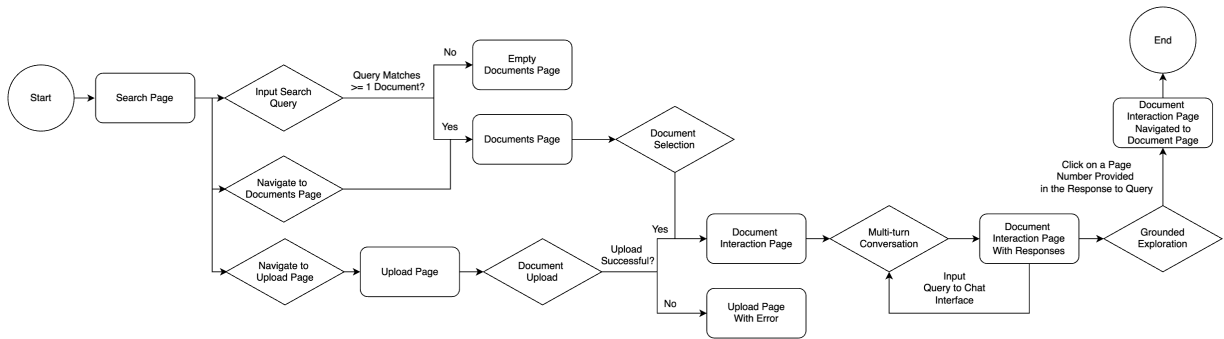


Figure 12: Proposed user flow based on the four use cases identified to target the user stories and objectives of our framework

6.2 Use Case 1: Semantic Search for Relevant Documents

The semantic search functionality plays a crucial role in enabling venture capital professionals to efficiently locate relevant documents based on their information needs. This use case leverages similarity searches to retrieve documents that are semantically related to the user's query, even in the absence of exact keyword matches.

6.2.1 FastAPI Route: Document Search and Interaction with Vector Database

The document search functionality is handled by the `/docs-search/{query}` route on the backend. This route receives the user's query as a parameter and initiates the process of retrieving relevant documents from the vector database. Upon receiving the query, the route handler interacts with the Qdrant vector database to find the most relevant document IDs through similarity searches. This process involves comparing the vector embedding of the user's query with the vector embeddings of the LlamaIndex Documents, which represent individual page contents in our framework. The vector database returns the document IDs with the highest similarity scores, indicating their relevance to the user's query. This approach allows the system to retrieve documents that are semantically similar to the query, even if they do not contain the same keywords.

After obtaining the relevant document IDs, the route handler retrieves the actual document data from the database, including the document name, date, and other relevant metadata. Finally, the route handler returns the retrieved documents as a JSON response to the frontend.

6.2.2 Semantic Search Frontend Implementation

The frontend implementation of the semantic search functionality revolves around providing an interface for users to enter their queries and initiate the search process. The search bar is prominently displayed on the frontend in the middle of the screen. It consists of an input field where users can type in their search queries (Figure 13).

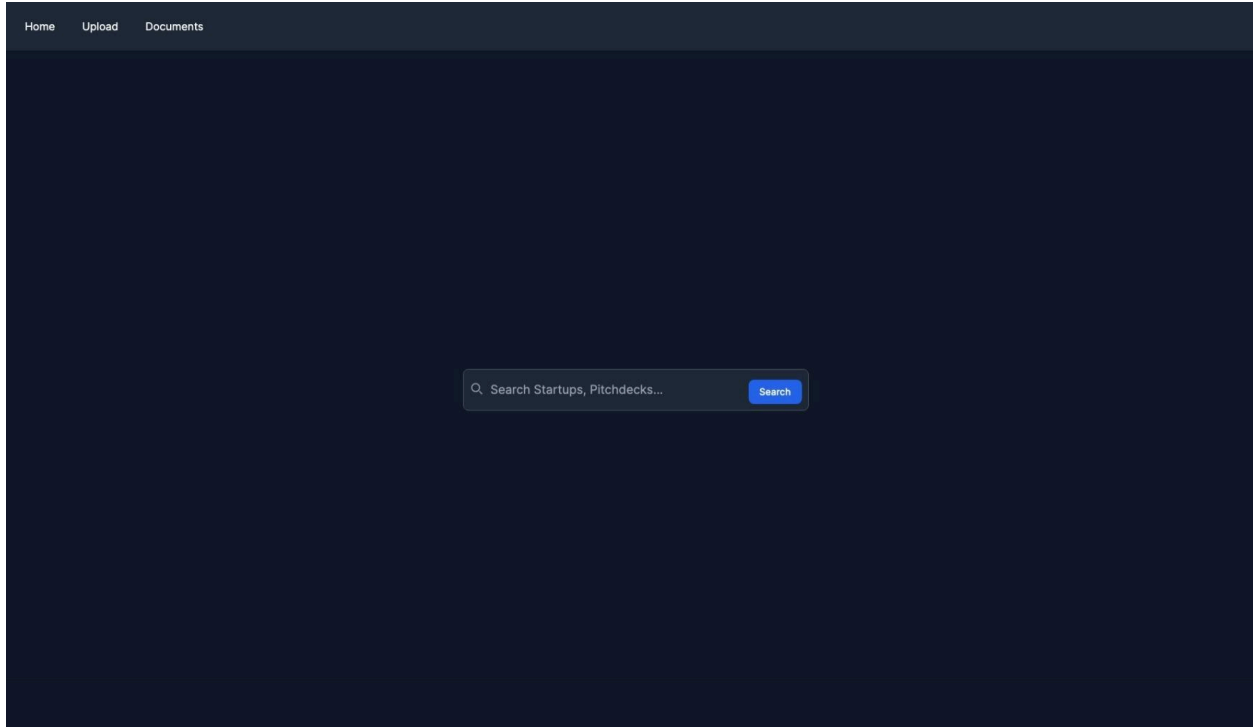


Figure 13: Frontend Interface for Semantic Search for Relevant Documents

6.3 Use Case 2: Document Selection

The document selection use case facilitates the efficient navigation and selection of relevant documents for further analysis. By providing a centralized view of all available documents, this use case empowers venture capital professionals to quickly identify and access the documents that align with their information needs.

6.3.1 FastAPI Route: Document Retrieval and Interaction with Relational Database

The document retrieval functionality is handled by the `/docs` route on the backend. This route serves as the entry point for fetching all available documents from the PostgreSQL relational database. When the frontend sends a request to the document retrieval endpoint, the route handler interacts with the PostgreSQL database instance. It invokes the appropriate methods to retrieve the list of documents stored in the relevant table. This process ensures that the backend can efficiently fetch the necessary document information, including document IDs, names, dates,

and other relevant metadata. Once the document data is retrieved from the database, the route handler formats the information into a JSON response and sends it back to the frontend.

6.3.2 Document Selection Frontend Implementation

Upon receiving the list of documents from the backend, the frontend renders them in a table view. The table view presents the documents in a structured and organized manner, with each row representing a single document. The columns of the table display relevant information about the documents, the document name and a button that allows users to seamlessly transition to the "Chat with Document" page (Figure 14), where they can engage in an interactive conversation with the selected document.

<removed>

Figure 14: Frontend Interface for Document Selection

6.4 Use Case 3: Document Upload

6.4.1 FastAPI Route: Document Upload

The document upload functionality is handled by a dedicated route on the backend. This route is responsible for receiving the uploaded document and triggering the necessary processing steps.

When a user selects a document to upload, the frontend sends a POST request to the /upload endpoint, including the document file in the request payload. The backend route handler receives the file and performs the necessary steps to store the file, create an instance of the relevant classes, generate a unique identifier (UID) for the upload task, and initiate a background task for processing the document. The background task, initiated by the route handler, processes the uploaded document using the appropriate classes and methods from the Extraction Phase. This process encompasses chart detection, information extraction, and storage of the extracted data in the respective databases (PostgreSQL and Qdrant).

6.4.2 Document Upload Frontend Implementation

The frontend implementation of the document upload functionality focuses on providing a user-friendly interface for selecting and uploading documents, as well as keeping the user informed about the progress of the upload and processing tasks.

The user interface includes an intuitive way for users to select the document file they wish to upload, such as an upload button or a drag-and-drop area (Figure 15). When a user selects a file, the frontend initiates the upload process by sending a POST request to the designated backend endpoint, including the selected file in the request payload. Upon receiving the response from the backend, which includes the document ID for the upload task, the frontend initiates the polling mechanism. It starts sending periodic requests to the backend endpoint responsible for providing the upload status, passing the document ID as a parameter to retrieve the status of the upload and processing task. The frontend updates the user interface based on the received status, providing visual feedback to the user about the progress of the task. This feedback can be in the form of progress indicators, loading animations, or informative messages that convey the current stage of the process.

Once the frontend receives a status indicating that the processing is complete, it automatically redirects the user to the page displaying the newly uploaded document corresponding to the document ID. This page corresponds to the designated path or route for viewing individual documents (`docs/{doc_id}`), allowing the user to seamlessly transition from the upload process to interacting with the document itself.

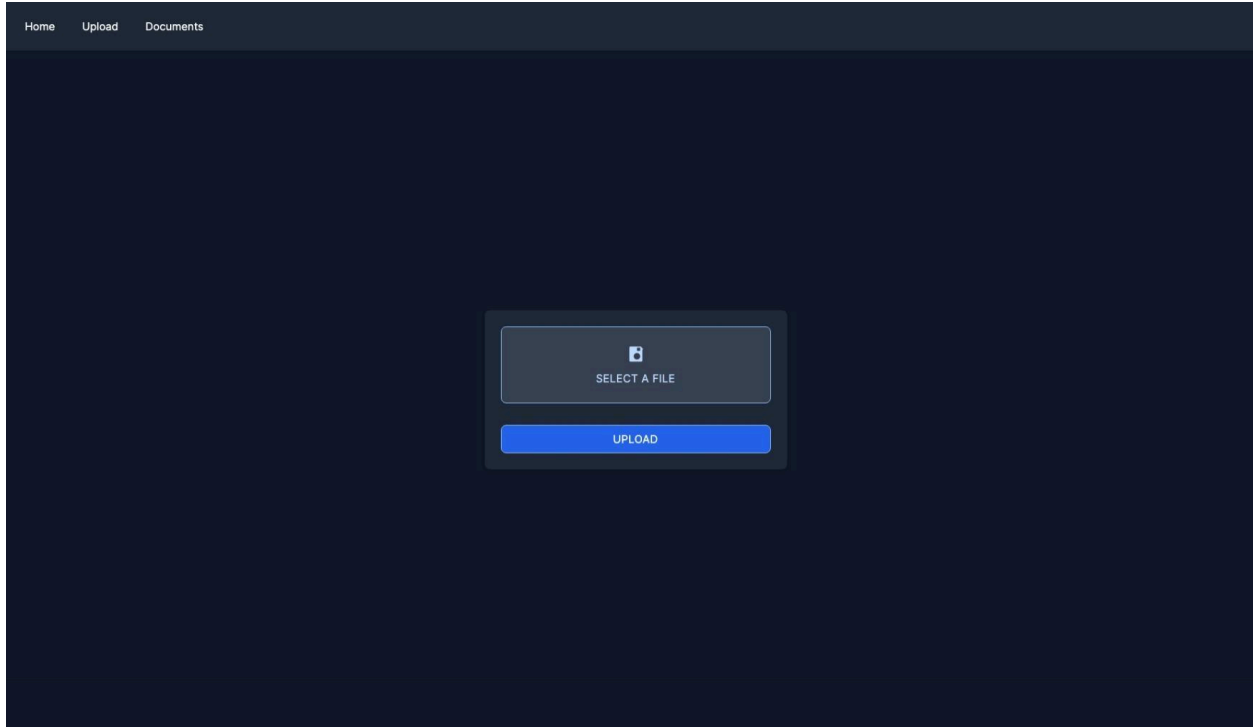


Figure 15: Frontend Interface for Document Upload

6.5 Use Case 4: Chat with Document

The "Chat with Document" use case represents the culmination of our framework, where venture capital professionals can engage in interactive, context-aware conversations with individual pitch deck documents. This use case leverages the accomplishments of the previous phases, seamlessly integrating the extracted information and enabling users to obtain insightful responses tailored to their queries.

6.5.1 FastAPI Routes: Document Information and PDF Rendering

To render pitch deck document-specific information on the frontend, the backend provisions routes specific to the document. The `/docs/{doc_id}` route is called to retrieve information from the PostgreSQL database. Specifically, metadata such as the name of the document and the locations of bounding boxes of its chart instances alongside their pages is retrieved (Appendix D, Figure D.2). This allows the frontend to render such information to the user for grounded exploration.

The backend also provisions a `/get-pdf/{doc-id}` that fetches a buffer containing the PDF of the pitch deck document retrieved locally based on the path of the document. This allows the frontend to display the PDF document alongside a chat interface with which the user can interact with.

6.5.2 FastAPI Route: Chat

Building upon the indexing step accomplished in the Extraction Phase, the implementation of the chat functionality completes the Retrieval-Augmented Generation (RAG) pipeline. The retrieval and generation steps of RAG are encapsulated within the `/ask` route on the backend (Figure 16).

The retrieval and generation steps of RAG are handled by the `/ask` route on the backend. In the retrieval step, when a user submits a query through the chat interface, this route receives the document ID and the conversation history. The latest user message is then extracted from the conversation history and retrieves the relevant context from the vector database using a specialised method. This method applies metadata filters to constrain the search results to the specified document ID, ensuring that only the relevant information from the selected document is considered during the retrieval process. By utilising metadata filters, the search space is narrowed down, leading to faster query times and more targeted results.

In the generation step, the retrieved context is integrated into the human message, transforming the query through augmentation with context. This provides the LLM with a comprehensive knowledge base specific to the context. The modified message includes instructions for the language model to refer to the context, answer the question in detail, provide insightful explanations, use the context as much as possible, quote relevant parts of the context to support the answer, and include the page number(s) where the relevant information is found.

For the LLM, we utilise Anthropic’s `claude-3-opus-20240229` model, also known as Claude Opus (Anthropic, 2024). As of the time of submission of the dissertation, Opus tops the LMSYS Chatbot Arena (Zheng et al., 2023), a crowdsourced open platform for LLM evaluations. We also experimented with LLMs on par with Opus, such as GPT-4, and found Opus’ responses to be more stable and insightful. While this assessment was qualitative, we propose to evaluate our framework with a more quantitative approach in Sections 7 and 8.

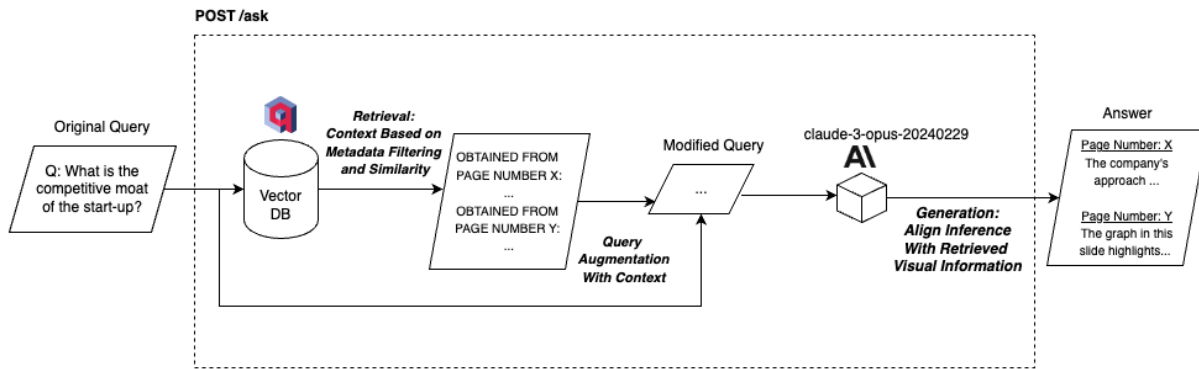


Figure 16: Showing the retrieval and generation steps of RAG encapsulated within the /ask route in the FastAPI backend

Lastly, the /ask route (Figure 16) utilizes a streaming method to generate a response based on the formatted conversation history. The LLM generates the response incrementally, allowing for real-time streaming of the generated content to the user. This streaming approach ensures a smooth and interactive user experience, as users can see the response being generated in real time.

6.5.3 Chat with Document Frontend Implementation

The frontend implementation of the chat with document use case focuses on providing a seamless multi-turn conversational experience. Users can engage in multiple rounds of interaction with the document, asking follow-up questions, seeking clarifications, and exploring different aspects of the document. The chat interface maintains the conversation history, allowing users to refer to previous interactions and maintain context throughout the conversation.

<removed>

Figure 17: Frontend Interface for Chat with Document showing grounded exploration and chat interaction

To enhance the user experience, the frontend implements real-time streaming of the generated responses. As the language model generates the response incrementally, the frontend dynamically updates the chat interface, displaying the response as it is being generated. This

streaming approach creates a more engaging and interactive experience, as users can observe the response unfolding in real-time.

One of the key features of this use case is the grounded reasoning aspect facilitated by the Retrieval-Augmented Generation (RAG) pipeline. The generated responses are grounded in the relevant parts of the document, with references to specific page numbers and bounding boxes of the supporting charts or content (Figure 17). This grounding mechanism enhances user experience by providing explainability and transparency in the generated insights.

The chat interface seamlessly integrates the accomplishments of the detection and extraction phases, allowing users to interact with the document conversationally and obtain insights grounded in the actual content. Users can click on the referenced page numbers or bounding boxes to navigate directly to the corresponding sections (Figure 18), strengthening the credibility and understanding of the responses.

<removed>

Figure 18: Frontend Interface for Chat with Document showing a multi-turn conversation

6.6 Deployment Architecture and Infrastructure

To ensure the framework can be effectively utilized by venture capital professionals in real-world scenarios and to facilitate ease of testing with users while simulating a production environment, careful consideration must be given to the deployment architecture, infrastructure, and ability to scale to handle increased usage.

Our current prototype leverages scalable and managed services to ensure efficient deployment and operation (Figure 19). The backend FastAPI service is deployed using Google Cloud Run, enabling automatic scaling based on incoming requests. The frontend Next.js application is deployed with Vercel, providing seamless integration and optimized performance. Neon DB and Qdrant Cloud are used for hosting the PostgreSQL and Qdrant databases, respectively, offering scalable and reliable database solutions. Google Cloud Storage (GCS) serves as the object store for efficient document storage and retrieval.

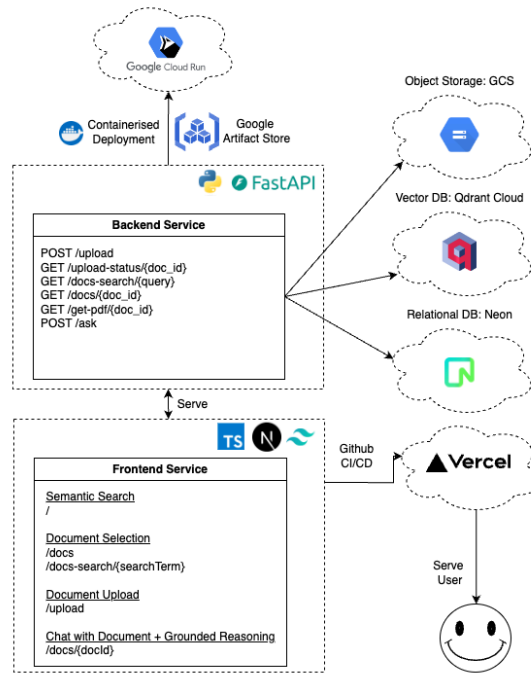


Figure 19: Deployment architecture for framework prototype

The prototype lays a solid foundation for scalability, utilizing serverless and managed services that automatically handle increased usage demands. Cloud Run ensures the backend service scales transparently, while Neon DB and Qdrant Cloud provide scalable database infrastructure. GCS offers high durability and scalability for document storage. The current prototype serves as a good starting point, demonstrating the feasibility and potential of the framework. As the user base expands and requirements evolve, the architecture can be further refined and optimized. Continuous monitoring, iterations, and adaptations will ensure the framework remains scalable, reliable, and efficient in production environments.

6.7 Completion of Synthesis Phase

The Synthesis Phase marks a significant milestone in the development of our end-to-end framework for knowledge retrieval from visually rich documents in the venture capital domain. This phase successfully brings together the achievements and contributions of the Detection and Extraction Phases, delivering a user-centric solution that addresses the key challenges faced by venture capital professionals in the deal-sourcing process.

The primary focus of the Synthesis Phase was to synthesise the extracted information and present it through an intuitive web application, empowering venture capital professionals with efficient access to relevant insights. By leveraging the Retrieval-Augmented Generation (RAG) pipeline, we have developed a powerful solution that enables users to engage in multi-turn conversations, ask specific questions, and obtain relevant insights grounded in the actual content of the documents.

One of the key achievements of the Synthesis Phase is the seamless integration of the backend routes and frontend implementations to support the identified use cases. The semantic search functionality (`/docs-search/{query}`) allows users to find relevant documents based on their meaning and context, while the document retrieval route (`/docs`) enables efficient access to the list of available documents. The document upload functionality (`/upload`) provides a user-friendly interface for uploading and processing new documents, while the chat with document use case (`/ask`) leverages the RAG pipeline to generate insightful responses grounded in the document content.

The frontend implementation of the Synthesis Phase plays a crucial role in delivering an intuitive and user-friendly interface. The semantic search bar, document selection table, document upload interface, and chat interface have been designed to enhance the user experience and facilitate seamless interaction with the framework. The real-time streaming of generated responses and the grounded reasoning aspect further elevate the user experience, providing explainability and transparency in the generated insights.

The Synthesis Phase directly addresses the user stories and objectives outlined in the Introduction (Section 1). It enables venture capital analysts to quickly obtain answers to frequently asked questions (FAQs) related to the competitive landscape and financial metrics of a start-up, streamlining the evaluation of investment opportunities. Moreover, it provides venture capital professionals with an intuitive interface to access synthesised information, navigate through processed documents, ask questions, and obtain relevant insights.

The deployment architecture and infrastructure considerations in the Synthesis Phase ensure the framework's scalability and reliability in real-world scenarios. By leveraging scalable and managed services such as Google Cloud Run, Vercel, Neon DB, Qdrant Cloud, and Google

Cloud Storage, the prototype demonstrates the feasibility and potential of the framework. The architecture is designed to handle increased usage demands and can be further refined and optimized as the user base expands and requirements evolve.

In conclusion, the Synthesis Phase marks the successful completion of our end-to-end framework, delivering a user-centric solution that addresses the challenges faced by venture capital professionals in the deal-sourcing process. By synthesising the extracted information, providing an intuitive web application, and leveraging the RAG pipeline, we have empowered users to access relevant insights efficiently and make informed investment decisions. The Synthesis Phase showcases the effectiveness of the overall framework and highlights its potential to streamline the deal-sourcing process in the venture capital domain.

7 Evaluation of Framework

7.1 Limitations on Current Benchmarks

Evaluating our framework poses challenges due to the lack of ground truth answers for the FAQs in each document. While publicly available benchmarks offer a potential solution, they are not well-suited to our specific use case. Question-answering benchmarks like ChartQA (Masry et al., 2022) and PlotQA (Methani et al., 2020) focus on individual chart instances rather than entire documents. Similarly, broad-based document understanding benchmarks such as DocVQA (Mathew et al., 2021) do not adequately capture the nuances of our framework, which targets pitch deck documents and emphasizes chart extraction. To address these limitations, we propose a hybrid approach to benchmarking that is tailored to our framework's unique characteristics.

7.2 A Hybrid Approach to Benchmarking

Our hybrid benchmarking approach consists of two components: evaluating the relevancy of answers using RAGAS and assessing the correctness of answers through surveys with venture capital professionals.

7.2.1 Evaluating Relevance with RAGAS

We employ the context relevancy metric (Appendix F) from RAGAS: Automated Evaluation of Retrieval Augmented Generation (Es et al., 2023). The metric assesses the quality of the retrieved information and its impact on the relevancy of the generated responses. We hypothesise that the Detection and Extraction Phases, which focus on extracting information from charts to answer the FAQs, significantly contribute to the relevancy of the responses.

To test this hypothesis, we compare two versions of our framework: the complete framework, including the Detection and Extraction Phases, and a baseline version that omits these phases and relies solely on the Synthesis Phase. We create separate vector database collections for each version and apply the Synthesis Phase to obtain answers for the nine FAQs. The evaluation score for each system is calculated as the aggregated answer relevancy across the FAQs for a single document, as shown in Equation 1. The intuition behind Context Relevancy is that if the generated answer sufficiently addresses the initial question, a language model should be able to generate questions from the answer that align with the original question (Es et al., 2023).

$$aggregated\ relevancy\ score = \frac{1}{KN} \sum_{j=1}^K \sum_{i=1}^N \cos(E_{g_{ij}}, E_o)$$

Equation 1: Context Relevancy over the FAQs for a Single Pitch Deck Document

Where:

E_{g_i} is the embedding of the generated question i , based on our generated answer.

E_o is the embedding of the original question.

N is the number of generated questions, which defaults to 3.

K is the number of FAQs, which is 9 in this case.

We use a simple mean across the nine FAQs because we found that outlier-immune measures such as the median heavily penalise instances of zero relevancy. Experiments revealed that certain questions are likely to be unanswerable, meaning that the answer is not available for the pitch deck. We hypothesize that this occurs when both systems yield zero relevancy for a question. Using a median in this case risks encountering zero medians often across systems for

documents, making it challenging to benchmark the systems together. Thus, a simple mean allows for less penalization of zero-relevancy scores, enabling comparison between the two systems.

The evaluation is repeated on a random sample of documents from the OSV pitch decks dataset (Figure 20).

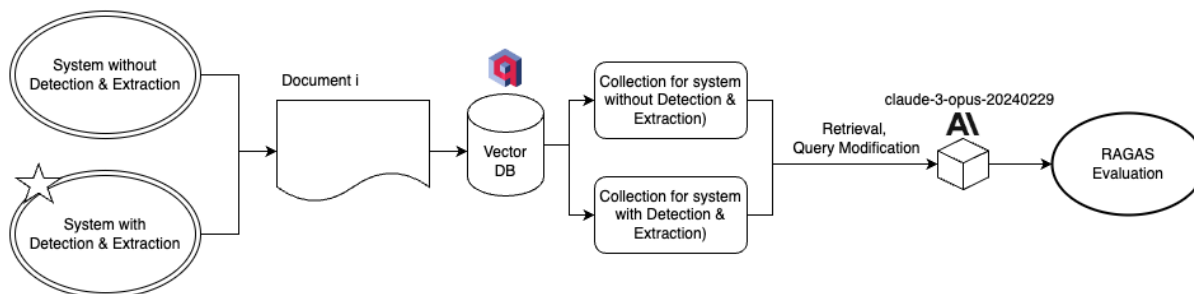


Figure 20: Workflow for RAGAS-based evaluation methodology on the baseline and complete systems

7.2.2 Evaluating Correctness through Surveys with Venture Capital Professionals

While RAGAS provides a robust assessment of the relevancy of the generated responses, it does not directly measure their correctness. To comprehensively evaluate the correctness of the responses, we propose conducting surveys with venture capital professionals. This approach involves selecting a representative sample of pitch decks from our dataset and generating responses to the FAQs using our framework. The generated responses, along with the corresponding pitch decks, will be presented to a panel of experienced venture capital professionals. These domain experts will be asked to rate the correctness of each response based on their expertise and the information provided in the pitch decks.

The surveys will be designed to gather both quantitative and qualitative feedback. Quantitative ratings will enable us to calculate metrics such as average correctness scores and inter-rater agreement, providing a standardized measure of the framework's performance. Qualitative feedback, on the other hand, will offer invaluable insights into the strengths and weaknesses of

our framework, helping us identify areas for improvement and guiding future development efforts. The full survey questionnaire can be found in Appendix D.

It is important to acknowledge that due to scheduling constraints, we were unable to obtain a comprehensive coverage of survey responses within the scope of this dissertation. However, this limitation sets the foundation for future work, where an even more comprehensive benchmark can be developed. The creation of a benchmark specifically tailored to the venture capital domain will enable a more rigorous evaluation of the framework's correctness and provide a valuable resource for the research community. Further details on this future direction will be discussed in the upcoming section.

Despite the absence of a complete correctness evaluation at this stage, it is crucial to emphasize that relevancy, as measured by RAGAS, serves as a strong indicator of the framework's effectiveness. The relevancy metric captures the essence of the framework's ability to extract and synthesise relevant information from pitch decks, which is a critical factor in its practical utility.

7.3 Evaluation Results

We present the results of our hybrid evaluation approach, focusing on the RAGAS-based assessment of relevancy. The evaluation is conducted on a sample of 132 pitch deck documents from the OSV dataset, comparing the performance of the complete framework against a baseline version without the Detection and Extraction Phases.

The results strongly support our hypothesis, with the complete system outperforming the baseline in 79.5% of the documents based on the Aggregated Relevancy Score. Furthermore, the boxplots in Figure 21 show that the complete system achieves a higher median score and exhibits greater consistency in generating relevant responses compared to the baseline.

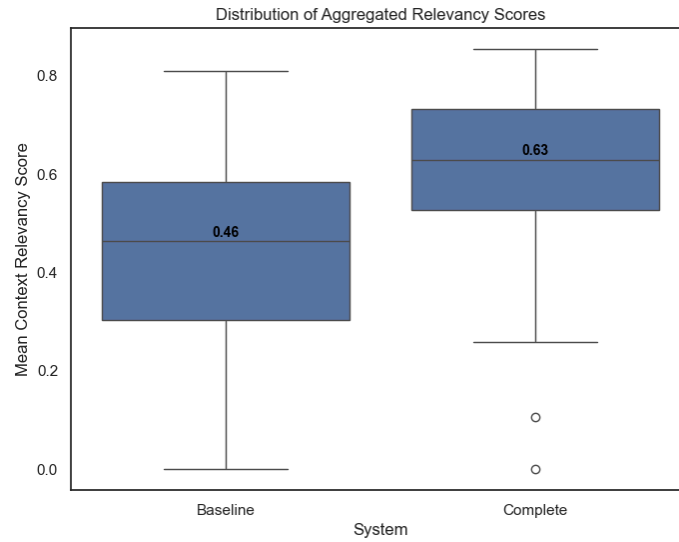


Figure 21: Boxplots of aggregated relevancy scores in the baseline and complete systems

Figure 22 presents a histogram of the score differences between the complete and baseline systems. The distribution is approximately normal, centred around the 0.1-0.2 range. Positive differences indicate that the complete system generally outperforms the baseline. A paired t-test confirms that the differences are statistically significant at a 5% level of significance.

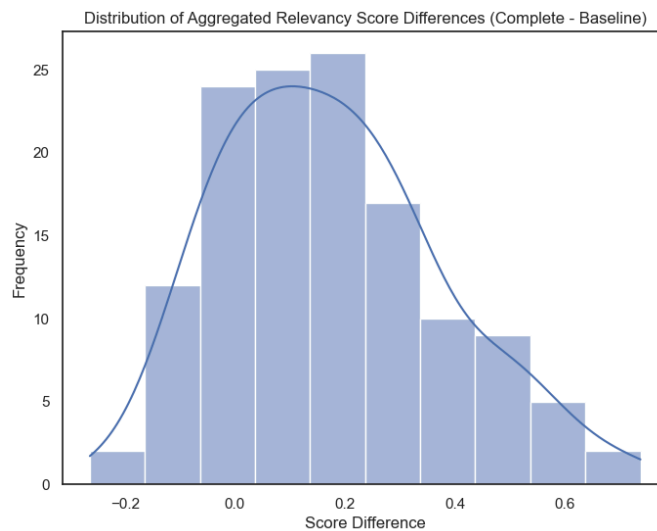


Figure 22: Histogram showing the distribution of score differences between the complete and baseline systems

Table 9 summarizes the system-level metrics, highlighting the superior performance of the complete system in terms of mean and median Aggregated Relevancy Scores. The baseline system's lower scores are attributed to its tendency to generate irrelevant responses for some documents.

Framework	Mean of Aggregated Relevancy Scores	Median of Aggregated Relevancy Scores	Standard Deviation of Aggregated Relevancy Scores
Baseline	0.435	0	0.427
Complete	0.609	0.768	0.357

Table 9: Table showing aggregations on a system level

It is important to note that the seemingly low average relevancy scores at the system level (0.435 and 0.609) do not necessarily indicate poor performance, as certain FAQs may be unanswerable based on the available information in the pitch decks. This limitation shows that there is a need for a more comprehensive evaluation framework that considers both relevancy and correctness.

Lastly, we assess the speed of the Detection and Extraction Phases, as shown in Table 10. The Extraction Phase emerges as a bottleneck, with an average time of 50.12 seconds, while the Detection Phase takes only 1.89 seconds on average.

Framework	Detection Time (s)	Extraction Time (s)	Average Total Time (s)
Baseline	-	-	0.02
Complete	1.89	50.12	52.01

Table 10: Table showing the speed of Detection and Extraction Phases

Despite the computational overhead introduced by the Detection and Extraction Phases, our evaluation demonstrates that the complete framework retrieves highly relevant responses to address FAQs in the deal-sourcing context. While further optimizations can be explored, the relevancy of the generated responses remains a key strength of our approach.

8. Conclusion

8.1. Summary

In this dissertation, we have presented a comprehensive end-to-end framework for knowledge retrieval from visually rich documents, with a specific emphasis on applications in the venture capital domain. The framework addresses the critical challenges faced by venture capital professionals during the deal-sourcing process, where the manual analysis of pitch decks and other visually rich documents can be time-consuming and error-prone.

One of the key achievements of this work is the development of a three-phase approach, comprising the Detection, Extraction, and Synthesis phases. In the Detection Phase, we made a significant contribution by creating the IIIT-OSV-Charts dataset, a novel dataset specifically curated for chart detection in the venture capital domain. This dataset combines the IIIT-AR-13K dataset with a proprietary dataset from Openspace Ventures, providing a robust foundation for training and evaluating state-of-the-art object detection models, such as YOLOV8-Nano and YOLOV8-Medium, to accurately identify and localize chart instances within pitch decks.

The Extraction Phase marks another notable achievement, where we successfully adapted the Set-of-Marks prompting strategy for grounded zero-shot understanding of charts within visually rich documents. By leveraging large multimodal models (LMMs) and utilizing the visual and textual context within the documents, we were able to extract relevant insights effectively. The extracted information was then efficiently stored in a vector database, enabling seamless retrieval and querying in the subsequent phases.

In the Synthesis Phase, we developed a sophisticated Retrieval-Augmented Generation (RAG) pipeline tailored to the deal-sourcing context. This pipeline synthesises the extracted information to generate comprehensive responses to frequently asked questions (FAQs) that are crucial for decision-making in the deal-sourcing process. The integration of the RAG pipeline with a user-friendly web application allows venture capital professionals to engage in multi-turn conversations and obtain relevant insights from the processed pitch decks, greatly enhancing their ability to make informed investment decisions.

Throughout the development of this framework, we have conducted a thorough evaluation using a hybrid approach, assessing the relevancy of the generated responses. The evaluation results demonstrate the effectiveness of our methodology in extracting relevant information from charts and generating accurate and insightful responses to FAQs, validating the value of our contributions.

In conclusion, the end-to-end framework presented in this dissertation represents a significant step forward in automating the analysis of visually rich documents, with a primary focus on the venture capital domain. However, the potential applications of this framework extend far beyond venture capital. The key achievements, including the creation of a novel dataset, the adaptation of advanced prompting strategies, and the development of a tailored RAG pipeline, have broad implications for industries and use cases that rely heavily on chart-based information.

Industries such as finance, healthcare, and business consulting often deal with large volumes of visually rich documents containing charts, graphs, and other data visualizations. The proposed framework can be adapted to streamline the analysis and extraction of insights from these documents, enabling professionals in these fields to make data-driven decisions more efficiently. The framework can be customized to address the specific needs and challenges of each industry.

8.2 Limitations

8.1.1 Reliance on Closed-source Models

One of the major limitations of this research is the heavy reliance on closed-source models from providers such as OpenAI and Anthropic. This dependence poses significant risks in terms of reliability and scalability. As highlighted in the Evaluation section, the Extraction Phase, which involves sending pages with charts to these closed-source models, becomes a bottleneck in the development process. The limitations are two-fold: firstly, the scope for code optimizations is limited since the infrastructure is owned by these companies; secondly, relying on external providers introduces a concentration risk, where any downtime in the model serving by these companies can severely impair the operations of our framework. To mitigate this concentration risk and ensure a more robust system, it is crucial to develop strategies that reduce the dependence on closed-source models and explore alternative solutions.

8.1.2 Uncertainty in Model Outputs

Uncertainty in model outputs is another significant limitation that can impact the entire framework in terms of correctness and robustness. This uncertainty manifests in different phases of the framework. In the Detection Phase, false positives, where charts are incorrectly identified, can confuse the Language Model (LMM) in the downstream tasks, especially when provided with a prompt specifically targeted at extracting information from charts. False negatives, where charts are present but not identified, can compromise the quality of extraction in subsequent phases. In the Extraction Phase, even though the adapted Set-of-Marks prompting allows for stronger reasoning capabilities and reduces the risk of hallucination, there is still an inherent risk when passing images of pages with charts to the LMM.

It is worth noting that in the Synthesis Phase, there is a chance of model hallucination when answering FAQs based on contextual information supplied by the vector database. Although the final generation is augmented with retrieved contextual information based on queries, there is still a small risk of model hallucination. Moreover, the uncertainties in the upstream phases, namely the Detection and Extraction Phases, can propagate and impact the final correctness of the output. Therefore, it is crucial to address all potential fault lines, starting from the Detection Phase, to minimize the impact of uncertainty on the overall framework.

8.1.3 Lack of a Rigorous Benchmark to Test Framework

As discussed in Section 7.2.3, we proposed and utilized a hybrid framework for evaluation. Although the results are promising, the evaluation does not comprehensively assess the framework's performance. Developing a novel benchmark specifically tailored to the venture capital domain would provide a more rigorous and reliable way to evaluate the framework. Such a benchmark should encompass a diverse set of pitch decks, ground truth answers for the FAQs, and well-defined metrics to measure the framework's performance regarding relevancy, correctness, and efficiency. Creating a standardized benchmark would not only enable a more thorough evaluation of our framework but also facilitate comparisons with future research and alternative approaches in this domain.

8.3 Recommendations for Future Work

8.2.1 Controlling Model Deployments through Employment of Open-Source Models

To mitigate the risks associated with relying on closed-source models, future work should explore the development of in-house LMMs and LLMs. By creating these models within the organization, we can gain significantly more control over the computing infrastructure, model fine-tuning, and architectural optimizations. This approach would reduce the dependence on external providers and minimize the impact of any potential downtime or limitations imposed by closed-source models.

However, it is important to acknowledge that developing in-house models requires substantial resources in terms of model management and infrastructure. A comprehensive cost-benefit analysis should be conducted to assess the feasibility and long-term sustainability of this approach, especially in the context of venture capital, where resources may be limited and speed is of the essence. Future work should carefully evaluate the trade-offs between the benefits of increased control and the associated costs and resource requirements.

8.2.2 Mitigating Uncertainty in Model Outputs

8.2.2.1 Mitigating Uncertainty in the Detection Phase with Model Retraining

To address uncertainty in the Detection Phase, future work should focus on implementing a model retraining pipeline that enables continuous improvement of the chart detection model. By adapting the web application to allow users to add and edit bounding box labels for chart instances, we can create a feedback loop that facilitates the collection of targeted annotations.

Upon receiving new or updated labels, the system can trigger model retraining in batches, gradually improving the performance of the chart detection model over time. As the model becomes more accurate and reliable, the uncertainty in the Detection Phase will be significantly reduced. Furthermore, the improved detection results can be propagated to the downstream phases, reducing uncertainty throughout the entire framework.

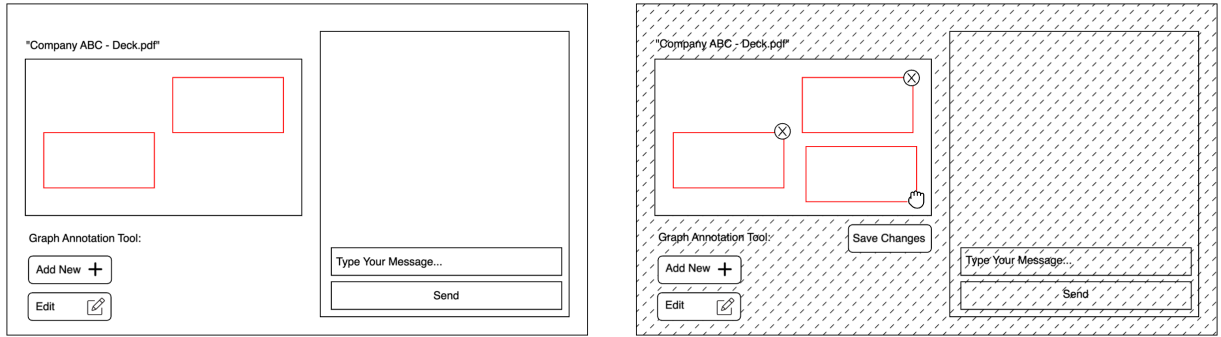


Figure 23: low fidelity mock-up demonstrating continuous learning

Figure 23 presents a low-fidelity mock-up demonstrating the concept of continuous learning in the context of our web application. The "Chart Annotation Tools" can be integrated into the "chat with document" page, enabling users to add new bounding boxes or edit existing ones. The user can commit changes by clicking on "Save Changes," which will update the annotations stored in the Postgres relational database. To fully realize this continuous learning pipeline, future work should explore the implementation of an MLOps pipeline on a platform like Google Cloud Platform. A Cron job (Cloud Scheduler) can be set up to schedule periodic retraining of the YOLOV8-Nano model, abstracting the model serving from the FastAPI backend service. The modular architecture of our system makes such enhancements highly feasible and achievable in future iterations (Figure 24).

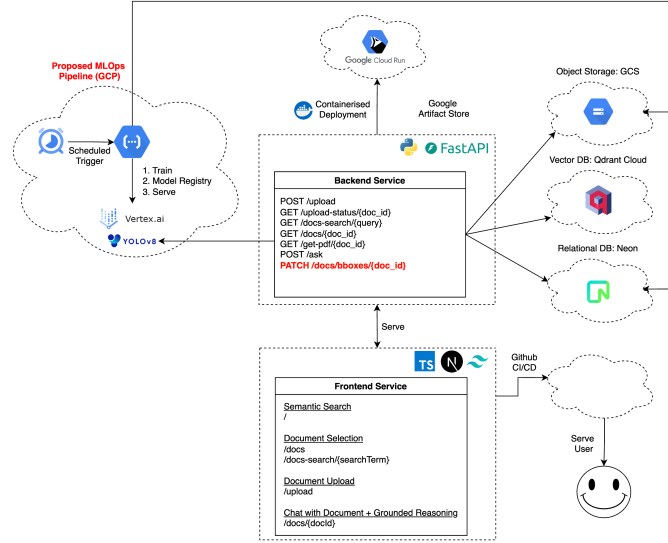


Figure 24: Proposed architecture changes incorporating MLOps

8.2.2.2 Mitigating Uncertainty in the Extraction and Synthesis Phases

To mitigate uncertainty in the Extraction and Synthesis Phases, future work should focus on continuous benchmarking and iterative improvements. By adapting the hybrid evaluation framework proposed in Section 7.2, we can systematically assess the performance of the framework and identify areas for optimization.

In the context of retrieval relevancy, future work can explore experimenting with different embedding models, chunking strategies, and LMMs to identify the most effective combinations. By comparing the results obtained from various configurations, we can tune the Synthesis Phase to achieve better relevancy scores. To address the correctness of outputs, future work should prioritize the development of a novel benchmark specifically designed for the venture capital deal-sourcing context. This benchmark, which will be discussed further in Section 8.3.3, will provide a standardized means to evaluate the correctness of the generated responses.

By continuously evaluating the framework against this benchmark and making iterative improvements, we can progressively enhance the relevancy and correctness of the outputs, thereby mitigating uncertainty over time. Moreover, the process of benchmarking and improvement can be automated, similar to the model retraining pipeline discussed in Section 8.3.2.1, to ensure a systematic and efficient approach to uncertainty mitigation.

8.2.3 Creation of a Novel Benchmark Targeting Pitch Decks in a Deal-sourcing Context

While the proposed survey in Section 7.2.2 provides a valuable starting point for collecting preliminary feedback on the framework, future work should prioritize the development of a comprehensive benchmark specifically tailored to the venture capital deal-sourcing context.

Drawing inspiration from popular benchmarks like ChartQA (Masry et al., 2022) and PlotQA (Methani et al., 2020), we can conceptualize a benchmark by obtaining ground truth answers for the FAQs outlined in Section 1.1.3, using the Proprietary Openspace Ventures Dataset (OSV Pitch Deck Dataset) introduced in Section 3.1.1.2. This process would involve annotating each document with the correct answers to the FAQs and creating a rich dataset for evaluation.

Once the ground truth answers are established, the performance of our framework can be evaluated by comparing its generated responses with the ground truth. Adapting metrics from the RAGAS framework (Es et al., 2023), such as the answer correctness metric, which considers both factual consistency and semantic similarity, can provide a robust assessment of the framework's performance. Factual consistency, or faithfulness, can be measured using the Faithfulness Score (Appendix E), which quantifies the proportion of claims in the generated answer that can be inferred from the given context. Semantic similarity, on the other hand, can be calculated using cosine similarity between the vectorized ground truth answer and the generated answer, utilizing the same embedding model.

The choice of semantic similarity as a key measure of correctness is preferred over edit-distance-based metrics like the Average Normalized Levenshtein Similarity (ANLS) (Biten et al., 2019). This is because the FAQs in the venture capital context are highly nuanced and require in-depth analysis to arrive at an answer. While ANLS has its strengths in softly penalizing OCR mistakes, which aligns with our use case of zero-shot understanding with LMMs, a strict edit distance-based metric may heavily penalize the generated responses to our complex FAQs. Therefore, a semantic-based distance metric like the answer correctness proposed in RAGAS is more appropriate.

As the framework evolves and matures, future work may explore alternative ways to measure correctness, adapting to the specific needs and challenges encountered in the venture capital domain. To the best of our knowledge, the conceptualization of a benchmark targeting chart

understanding in the venture capital context would be the first of its kind. We strongly recommend that future research efforts be directed towards the development and refinement of this benchmark, as it will serve as a valuable resource for advancing the state-of-the-art in this domain.

References

- Anthropic. (2024). *Introducing the next generation of claude*.
<https://www.anthropic.com/news/claude-3-family>
- Belval. (n.d.). *Belval/pdf2image: A python module that wraps the PDFTOPPM utility to convert PDF to PIL image object*. GitHub. <https://github.com/Belval/pdf2image>
- Biten, A. F., Tito, R., Mafla, A., Gomez, L., Rusiñol, M., Valveny, E., Jawahar, C. V., & Karatzas, D. (2019, October 16). *Scene text visual question answering*. arXiv.org. <https://arxiv.org/abs/1905.13648>
- Chen, J., Xiao, S., Zhang, P., Luo, K., Lian, D., & Liu, Z. (2024, February 10). *BGE M3-embedding: Multi-lingual, multi-functionality, multi-granularity text embeddings through self-knowledge distillation*. arXiv.org. <https://arxiv.org/abs/2402.03216>
- Es, S., James, J., Espinosa-Anke, L., & Schockaert, S. (2023, September 26). *RAGAS: Automated Evaluation of retrieval augmented generation*. arXiv.org. <https://arxiv.org/abs/2309.15217>
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2021, April 12). *Retrieval-augmented generation for knowledge-intensive NLP tasks*. arXiv.org. <https://arxiv.org/abs/2005.11401>
- Li, J., Xu, Y., Lv, T., Cui, L., Zhang, C., & Wei, F. (2022, July 19). *DIT: Self-supervised pre-training for document image transformer*. arXiv.org. <http://arxiv.org/abs/2203.02378>
- Liu, F., Eisenschlos, J. M., Piccinno, F., Krichene, S., Pang, C., Lee, K., Joshi, M., Chen, W., Collier, N., & Altun, Y. (2023, May 23). *DePlot: One-shot visual language reasoning by plot-to-table translation*. arXiv.org. <http://arxiv.org/abs/2212.10505>
- LlamaIndex. (2024a). *Documents / Nodes*.
https://docs.llamaindex.ai/en/stable/module_guides/loading/documents_and_nodes/
- LlamaIndex. (2024b). *LlamaIndex, Data Framework for LLM applications*.
<https://www.llamaindex.ai/>
- LlamaIndex. (2024c). *Node parser usage pattern*.
https://docs.llamaindex.ai/en/stable/module_guides/loading/node_parsers/
- Masry, A., Do, X. L., Tan, J. Q., Joty, S., & Hoque, E. (n.d.). *Chartqa: A benchmark for question answering about charts with visual and logical reasoning*. ACL Anthology.
<https://aclanthology.org/2022.findings-acl.177>

- Mathew, M., Karatzas, D., & Jawahar, C. V. (2021, January 5). *DocVQA: A dataset for VQA on Document Images*. arXiv.org. <https://arxiv.org/abs/2007.00398>
- Methani, N., Ganguly, P., Khapra, M. M., & Kumar, P. (2020, February 1). *Plotqa: Reasoning over scientific plots*. arXiv.org. <http://arxiv.org/abs/1909.00997>
- Mondal, A., Lipps, P., & Jawahar, C. V. (2020, August 6). *IIIT-ar-13K: A new dataset for graphical object detection in documents*. arXiv.org. <https://arxiv.org/abs/2008.02569>
- Nvidia H100 for AI & ML Workloads: Cloud GPU platform. Paperspace. (2024). <https://www.paperspace.com/>
- Obeid, J., & Hoque, E. (2020, November 29). *Chart-to-text: Generating natural language descriptions for charts by adapting the Transformer model*. arXiv.org. <http://arxiv.org/abs/2010.09142>
- OpenAI. (2024). *Embeddings - OpenAI API*. OpenAI Embeddings. <https://platform.openai.com/docs/guides/embeddings>
- PostgreSQL. (2024, March 30). *PostgreSQL*. <https://www.postgresql.org/>
- Pymupdf. (n.d.). *Pymupdf/pymupdf: Pymupdf is a high performance python library for Data Extraction, analysis, Conversion & Manipulation of PDF (and other) documents*. GitHub. <https://github.com/pymupdf/PyMuPDF>
- qdrant. (n.d.). *Vector database*. Qdrant. <https://qdrant.tech/>
- Rastogi, R. (2023, September 20). *Papers explained 19: DIT*. Medium. <https://medium.com/dair-ai/papers-explained-19-dit-b6d6eccd8c4e>
- Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016, May 9). *You only look once: Unified, real-time object detection*. arXiv.org. <https://arxiv.org/abs/1506.02640>
- Roboflow. (2024). *Give your software the power to see objects in images and video*. <https://roboflow.com/>
- Starttech Ventures. (2022, June 10). *What is a pitch deck and who is it for?* <https://www.starttech.vc/library/what-is-a-pitch-deck/>
- Weights & Biases. (2024). *Weights & Biases: The AI developer platform*. <https://wandb.ai/site>
- Yang, J., Zhang, H., Li, F., Zou, X., Li, C., & Gao, J. (2023, November 6). *Set-of-mark prompting unleashes extraordinary visual grounding in GPT-4V*. arXiv.org. <https://arxiv.org/abs/2310.11441>

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., & Stoica, I. (2023, December 24). *Judging LLM-as-a-judge with MT-bench and Chatbot Arena*. arXiv.org. <https://arxiv.org/abs/2306.05685>

Zhong, X., Tang, J., & Yepes, A. J. (2019, August 16). *PubLayNet: Largest dataset ever for document layout analysis*. arXiv.org. <https://arxiv.org/abs/1908.07836>

Appendices

Appendix A: Interview Summaries and Findings

A.1 Summary of Interviews Conducted with Venture Capital Professionals

Question	Summary of Answer (Interviewee <u>underlined</u>)	Insight
What is your job and role in the VC?	Refer to Table 5	Not Applicable
2. Describe the general deal-sourcing process in the firm	<u>Analysts B & C</u> : Usually starts with looking at the pitch decks received... Then we would conduct our analyses based on the information found in the decks... We sometimes look at research papers to get an understanding of the industry... We also conduct interviews with founders, customers, and expert calls too... Then we will present all that information during the Deal Review Committee (DRC) meeting...	The deal-sourcing process can take a significant amount of time
What are some key information required during deal sourcing?	<u>Analyst C</u> : “In terms of company-specific information, we look at revenue projections, margin profiles, list of existing investors,	Common analyses include retrieving key financial metrics, existing investors, outlook

	industry outlook... Those are some of the common items that we look at” <u>Analyst C:</u> “We also compare ... in terms of their valuations, existing investors...”	
4. Pertaining key financial metrics or key start-up information, where are they usually located in a pitch deck?	<u>Analyst D:</u> “Usually slides (pitch decks) are designed such that information related to a certain subject can be found within the same slide (page)... Financial metrics are usually found in charts”	Key information and topics are contained within individual pages. This means that a page-level analysis is required in order to obtain key information
What are your pain points during deal sourcing in terms of process efficiency?	<u>Analyst A:</u> “It is not easy to conduct analysis on these new companies, like comparing between startups we’ve seen so far ... Data is scarce apart from information from the pitch decks they have sent us” <u>Analyst B:</u> “There are many decks and documents that we would refer to... To get one piece of information often requires us to look at multiple sources”	It is important to extract as much information from a pitch deck as possible, as alternative data sources are scarce Manual analysis of documents is a tedious process There is a need for an easier way to obtain relevant documents pertaining a query (one piece of information)

Table A.1.1: Summary of interviews conducted to better understanding deal-sourcing and its pain points

S/N	Stage	Source Type	Document Name	Modes	Description
1	Deal-sourcing	Primary	Pitch Deck (PDF)	Image, Text	Created by startups. For VC analysts to refer to them while researching and analyzing in preparation for DRC. Contains basic information about the company, such as the founders, a business overview of the market, and technology.

2	Deal-sourcing	Primary	Research Reports (PDF)	Image, Text	For VC analysts to refer to them while researching and analysing in preparation for DRC.
3	Deal-sourcing	Secondary	Interview Notes (Word)	Text	For VC analysts to refer to them while researching and analysing in preparation for deal review meetings.
4	Deal-sourcing	Secondary	Internal Interim Documents (Word, PDF, Excel)	Depends	Documents related to ad hoc analyses performed in preparation for deal review meetings. Examples include returns analysis, and financial modelling

Table A.1.2: Findings from Interviews to find out key documents involved in deal sourcing

Name (Anonymised)	Position	Role(s)
Analyst A	Senior Analyst	Sources for new companies (deal sourcing) Focus on mid-to-late-stage start-ups
Analyst B	Analyst	In charge of investor relations Although is not involved in deal sourcing, he/she needs to gather information from investments (company-specific information) to convey information to the investors
Analyst C	Associate	Sources for new companies (deal sourcing) Focus mostly on existing portfolio companies, and early-stage start-ups
Analyst D	Director	Works with portfolio companies Although is not involved in deal sourcing, he/she needs to gather information from investments (company-specific information)

Table A1.2.3: Profiles of Interviewees

Appendix B: Data Collection and Processing

B.1 Data Processing Steps and Outcomes

Algorithm 1 Pitch Deck File Selection Algorithm

```
1: Initialize whitelist with keywords associated with pitch decks
2: Initialize blacklist with irrelevant keywords and exact terms
3: Define minimum file size threshold
4: for each file in the dataset do
5:   if file size  $\geq$  minimum threshold then
6:     Continue to the next file
7:   end if
8:   if file name starts with a six-digit number then
9:     Continue to the next file
10:  end if
11:  Initialize flag as potential pitch deck to False
12:  for each keyword in whitelist do
13:    if keyword found in file then
14:      Set flag as potential pitch deck to True
15:      Break
16:    end if
17:  end for
18:  if flag is True then
19:    for each term in blacklist do
20:      if term found in file then
21:        Set flag as potential pitch deck to False
22:        Break
23:      end if
24:    end for
25:  end if
26:  if flag is True then
27:    Add file to final selection list
28:  end if
29: end for
30: Exclude any files in the final selection that also appear in the exclusion list
31: return final list of potential pitch decks
```

Figure B.1 Pitch Deck File Selection Algorithm

B.2 Details of the IIIT-AR-13K Dataset

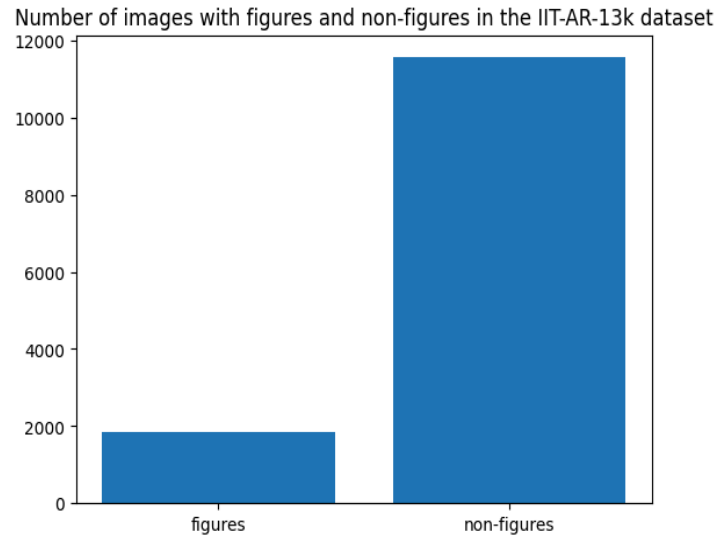


Figure B.2.1: Distribution of images from IIT-AR-13K based on whether they have figures

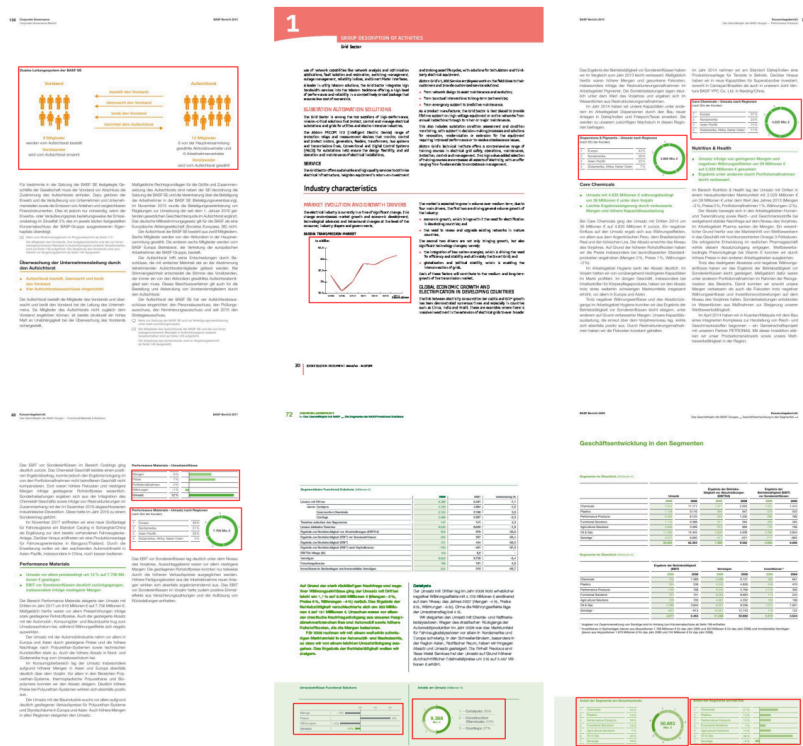


Figure B.2.2: 3x3 grid of images sampled from IIT-AR-13K, pre-annotated with bounding boxes representing figures

Appendix C: Chart Detection Model Training and Evaluation

C.1 Evaluation Metrics and Results

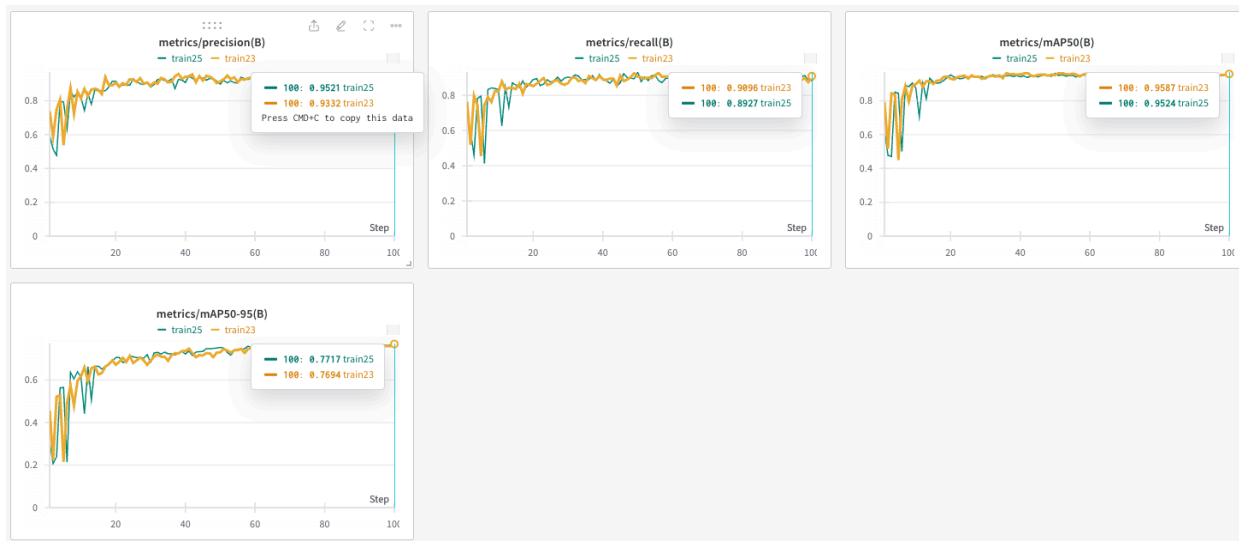


Figure C.1.1: Metrics during training over 100 epochs, train23 representing YOLOV8-Nano training and train25 representing YOLOV8-Medium training.



Figure C.1.2: Training and validation losses over 100 epochs, train23 representing YOLOV8-Nano training and train25 representing YOLOV8-Medium training.

Appendix D: Extraction Phase Details

D.1 Adapted Set-of-Marks Strategy

This is a page from a pitchdeck for the document named {doc_name}. It may include details on the startup's name, product, business model, market size, competition, founders, traction, unit economics, P&L shape, investment scope, exit potential, and current investors list.

Please extract all relevant information from this page. Additionally, you will encounter one or more charts annotated with bounding boxes. Based on these charts, provide a detailed overview of the startup's documentation, covering a wide range of aspects essential for a comprehensive understanding of its potential and current status:

- ***Sector/Business Model**: Examine the startup's industry and business model. Consider its position within the broader ecosystem and any innovative strategies it employs.*
- ***Product & Value Proposition**: Detail the startup's products or services and the unique advantages they offer to customers. Discuss how these offerings differentiate in the market.*
- ***Competition/Moat**: Analyze the startup's competitive landscape, including proprietary technology, strategic partnerships, or established barriers to entry.*
- ***Founder Background**: Offer insights into the founders' expertise, past ventures, and their vision for the company. Emphasize any unique experiences or qualifications that support the startup's success.*
- ***Market Size**: Evaluate the market opportunity for the startup's offerings. Look at current market conditions and future growth potential.*
- ***Impact**: Assess the tangible and intangible impacts of the startup's products or services on its customers and the industry.*

- **Current Traction**: Summarize key performance indicators, such as user growth, financial metrics (revenue, profit, expenses), and market penetration for the current and previous years, assuming the current year is 2024.
 - **Unit Economics**: Discuss the startup's cost structure and revenue model. Provide insights into customer acquisition costs, retention costs, and the lifetime value of a customer.
 - **P&L Shape**: Describe the startup's financial health and profitability trends, including major cost drivers and revenue streams.
 - **Investment Scope**: Outline the startup's funding goals, including the capital being raised, intended use of funds, and any pre-money or post-money valuations.
 - **Exit Potential**: Analyze potential exit strategies for the startup, considering industry trends, historical acquisitions, and the current investment climate. Discuss potential acquirers and whether an IPO or acquisition is more likely.
 - **List of Current Investors**: Compile a list of current investors, their investment stages, and any strategic alignments or partnerships that may affect the startup's path.
- Link the charts to the rest of the page to construct your analysis comprehensively. If certain information is not explicitly presented or is ambiguous, omit it.

Listing D.1 Extraction Prompt used in Extraction Phase

D.2 Data Storage and Database Design

PitchDeckDocs
id TEXT PRIMARY KEY
name TEXT NOT NULL
pagesWithChartsIdx INTEGER[]
bboxes FLOAT[][]]
docWidth INTEGER
docHeight INTEGER

Figure D.2: PitchDeckDocs schema in PostgreSQL

Appendix E: Evaluation Framework and Metrics

E.1 RAGAS Context Relevancy Metric

Relevancy can be measured as:

$$\text{answer relevancy} = \frac{1}{N} \sum_{i=1}^N \cos(E_{g_i}, E_o)$$

Equation 1: Answer Relevancy

Where:

E_{g_i} is the embedding of the generated question i .

E_o is the embedding of the original question.

N is the number of generated questions, which defaults to 3.

Equation E.1: RAGAS Context Relevancy Metric

E.2 Metrics Related to Proposed Evaluation Framework

$$ANLS = \frac{1}{N} \sum_{i=0}^N (s(a_{ij}, o_{qi}))$$

$$s(a_{ij}, o_{qi}) = \begin{cases} 1 - NL(a_{ij}, o_{qi}) & \text{if } NL(a_{ij}, o_{qi}) < \tau \\ 0 & \text{if } NL(a_{ij}, o_{qi}) \geq \tau \end{cases}$$

Equation E.2.1: Average Normalized Levenstein Similarity (ANLS)

$$\text{Faithfulness score} = \frac{|\text{Number of claims in the generated answer that can be inferred from given context}|}{|\text{Total number of claims in the generated answer}|}$$

Equation E.2.2: RAGAS Faithfulness Score

E.3 Proposed Human Evaluation Framework (Survey)

(Please refer to https://docs.google.com/document/d/1j5JkHZ70kr4JaUa4ouRjlqRNzO5WqO3fS-N7axWi820/edit?usp=drive_link)

Appendix F: Additional Resources

F.1 Sample Pitch Deck Pages with Charts

<removed>

Figure F1.1: Sample pitch deck page showing bar charts with vague axes

<removed>

Figure F.1.2: Sample pitch deck page showing pie charts with no labels in their slices

<removed>

Figure F.1.3: Sample pitch deck page showing line chart with no point annotations

<removed>

Figure F.1.4: Sample pitch deck showing bar charts with vague axes

<removed>

Figure F.1.5: Sample pitch deck showing line charts with no point annotations